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A comprehensive review of the deep learning-based tumor analysis approaches in histopathological images: segmentation, classification and multi-learning tasks

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Abstract
Medical Imaging has become a vital technique that has been embraced in the diagnosis and treatment process of cancer. Histopathological slides, which microscopically examine the suspicious tissue, are considered the golden standard for tumor prognosis and diagnosis. This excellent performance caused a sudden and growing interest in digitizing these slides to generate Whole Slide Images (WSI). However, analyzing WSI is a very challenging task due to the multiple-resolution, large-scale nature of these images. Therefore, WSI-based Computer-Aided Diagnosis (CAD) analysis gains increasing attention as a secondary decision support tool to enhance healthcare by alleviating pathologists' workload and reducing misdiagnosis rates. Recent revolutionized deep learning techniques are promising and have the potential to achieve efficient automatic representation of WSI features in a data-driven manner. Thus, in this survey, we focus mainly on deep learning-based CAD systems in the context of tumor analysis in histopathological images, i.e., segmentation and classification of tumor regions. We present a visual taxonomy of deep learning approaches that provides a systematic structure to the vast number of diverse models proposed until now. We sought to identify challenges that face the automation of histopathological analysis, the commonly used public datasets, and evaluation metrics and discuss recent methodologies for addressing them through a systematic examination of presented deep solutions. The survey aims to highlight the existing gaps and limitations of the recent deep learning-based WSI approaches to explore the possible avenues for potential enhancements.

Keyword Computer-aided diagnosis · Digital pathology · Histopathological images · Medical image segmentation · Medical image classification · Whole-slide images

1 Introduction

According to the global cancer report [1] of the World Health Organization, breast, lung, and colorectal cancers are the leading causes of the spread of new cancer cases worldwide, with a rate of 11.7%, 11.4%, and 10%, respectively. In addition, lung cancer is the first leading deadly threat, with a rate of 18%, followed by colorectal and liver cancers, with rates of 9.4% and 8.3%, respectively.

A cancer diagnosis is traditionally made through symptoms analysis and clinical exams. When a tumor is suspected, a diagnosis procedure must be performed. These diagnosis procedures usually belong to two main categories, noninvasive and invasive diagnoses. Noninvasive diagnosis refers to screening the tumor area through different imaging techniques that capture different anatomical and functional details [2–4] like Computed Tomography (CT), Magnetic Resonance Imaging (MRI), or Positron Emission Tomography (PET). On the other hand, invasive diagnosis refers to surgical biopsies of the suspicious region. The histopathological analysis of this extracted tissue is considered the golden standard for cancer diagnosis and grading [5] since it is a 100% definite and accurate test to confirm if the tumor is cancerous (i.e., benign or malignant) and determine its stage. Because of the terrifyingly large number of cancer incidences worldwide with (19,292,789) new cancer cases in 2020 [1]; the

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surging awareness of the importance of early detection [6]; and the needed follow-up treatments, the number of performed biopsies has been growing exponentially. However, there is a shortage of qualified pathologists [7] with significant workloads [8] required to meet the clinical demands of developing an appropriate treatment plan for each patient based on their diagnostic stage.

Moreover, the manual interpretation of pathological slides, either under a microscope or viewed through a computer interface using their digital counterparts (WSI), is subjective, laborious, and time-consuming; it depends on the pathologists' technical and clinical experience, training, and workflow. Thus, given the large number of slides the pathologist needs to analyze, the pathologist's fatigue level [9] is also a critical factor in the WSI interpretation's subjectivity and reproducibility issue. This inter and intra-observer variability causes limited agreement and decision-making discrepancy, consequently increasing the risk of misdiagnosis [10]. Thus, automation of this diagnostic workflow becomes necessary to support pathologists in the decision-making process [11] to reduce their burden to avoid any possible misdiagnosis caused by subjective assessment. Therefore, histopathological-based CAD systems, fueled by the recent artificial intelligence methods, act as diagnostic tools that provide a second opinion to pathologists to interpret the slides and process this huge amount of data.

The impressive and astonishing performance of Deep Learning (DL) in many vision tasks and its ability of an end-to-end self-extraction of a complex hierarchy of features based on the given data [12] make the deep learning-based CAD systems offer robustness and good generalization ability [13]. In particular, deep learning-based WSI tumor segmentation and classification methods have thrived in the last few years based on the number of publications. Deep learning mainly tries to overcome the operator dependency issue by replacing the human handcrafted discriminative features that depend on pathologists' domain knowledge and skills with self-learned features with different levels of representations. Although these methods lack the prior domain bias, they sacrifice the interpretation of the learned features [13]; they, therefore, are considered black boxes. Deep learning succeeds in automatically retrieving discriminative features that represent the images, but it is difficult to explicitly extract meaningful information from these features [14]. To minimize the effects of this issue, some WSI researchers tried to achieve interpretability by using heatmaps [15], saliency maps [16], visual attention maps [17], or uncertainty maps [18] to highlight the discriminative information that leads to the prediction.

The histopathological analysis is a vast discipline that includes many tasks and applications, such as cell and nuclei segmentation, normal and tumor cell counting, tissue recognition, and cancer grading. Due to the importance and the recent broad interest in the deep learning contribution to the histopathological analysis, some recent related surveys and reviews are available in the literature. However, these surveys are either general that cover different tasks and applications of the histopathological analysis in brief; general that focus only on the histopathological systems that used a specific deep learning model; specific that discuss a single histopathological task; or organ-specific. Table 1 lists these surveys along with their contributions. Thus, we have bounded the discussion in this survey to the studies of segmentation, classification and multi-learning tumor analysis tasks. We aim to establish a strong foundation and a general understanding to guide interested beginners and expert researchers to better explore the field. We particularly seek to answer the following research questions to provide a vital understanding of the current stand of deep learning-based histopathological tumor analysis research in order to propose new avenues:

- What are the main challenges and difficulties facing digital histopathological image analysis?
- What is the recent state-of-the-art histopathological tumor analysis approaches that utilize deep learning as their base?
- What are the future directions to enhance the existing histopathological-based CAD systems?

The main contributions of this survey are four-fold:

(1) We conduct a comprehensive review and analysis of the latest state-of-the-art deep learning-based approaches for tumor analysis in histopathological images proposed to address three tasks: segmentation, classification, and multi-learning. Afterward, we provide a unique methodology-based categorization to organize the approaches in each task according to the adopted learning strategy: unsupervised, semi- and weakly supervised, supervised, transfer, or reinforcement learning.

(2) We present an overview of five famous and widely-used public datasets in the histopathological-based tumor analysis field, comparing their specifications to identify their characteristics and evaluation metrics to understand the examined approaches' descriptions better.

(3) We provide an overview and detailed summarization of the preprocessing, the data augmentation, the dataset, and the evaluation metric used in each of the examined approaches. Furthermore, we collect and report the performances of these reviewed approaches on the used datasets.

Table 1 Recent surveys in histopathological analysis and their contribution

Survey references	Year	Contribution
[19]	2021	General histopathological tasks, not only tumor-specific
[20]	2021	General histopathological tasks, not only tumor-specific
[21]	2020	General histopathological tasks, not only tumor-specific and concerned with a particular type of deep learning models: Generative adversarial-based systems
[22]	2022	General histopathological tasks, not only tumor specific and concerned with a particular type of deep learning models: Graph neural-based systems
[23]	2019	Specific task: nuclei segmentation task
[24]	2021	Specific task for a specific organ: prostate cancer detection

(4) We highlight the current state and the trending directions in this area and identify the potential avenues for future research.

The body of research in this area is growing rapidly; thus, we limit the examination and analysis in this survey and cover the period from 2018 to the latest. Related keywords to the topic and their synonyms were used as the search string in the article selection process. After that, a filtering and exclusion step was conducted to reduce the number of duplicate and irrelevant papers. Finally, a quality verification and assessment step was performed based on the content, contribution, and structure to be included in the survey. Descriptive statistics of chosen and investigated papers according to their published year, their examined organ, and their specific task are illustrated in Fig. 1.

This survey is organized as follows. Section 2 discusses the challenges that face the automation of the histopathological analysis field. The following sections overview the general pipeline of any analysis approach, starting with introducing data preprocessing techniques that can be applied to enhance the analysis performance in Sect. 3. Section 4 discusses the evaluation metrics used to measure the accuracy of the segmented tumor boundaries or the predicted classification labels. The commonly used histopathological public datasets are presented in Sect. 5. Section 6 categorizes and summarizes different tumor analysis tasks, such as the segmentation schemes (Sect. 6.1), the classification schemes (Sect. 6.2), and the multi-task schemes (Sect. 6.3). Section 7 discusses the characteristics, limitations, and significant research trends and directions adopted by the investigated works to shed the light upon the possible directions for future enhancements in computational pathology in general and tumor analysis in particular. Finally, the conclusion is outlined in Sect. 8. Figure 2 illustrates the remaining structure of the survey.

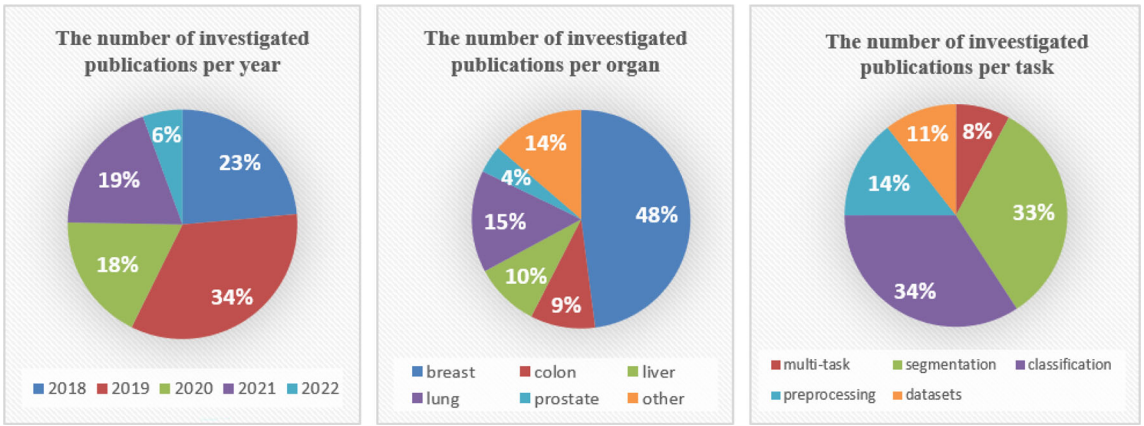


Fig. 1 Statistical distribution of investigated publications in this survey per; published year, examined organ, and targeted task

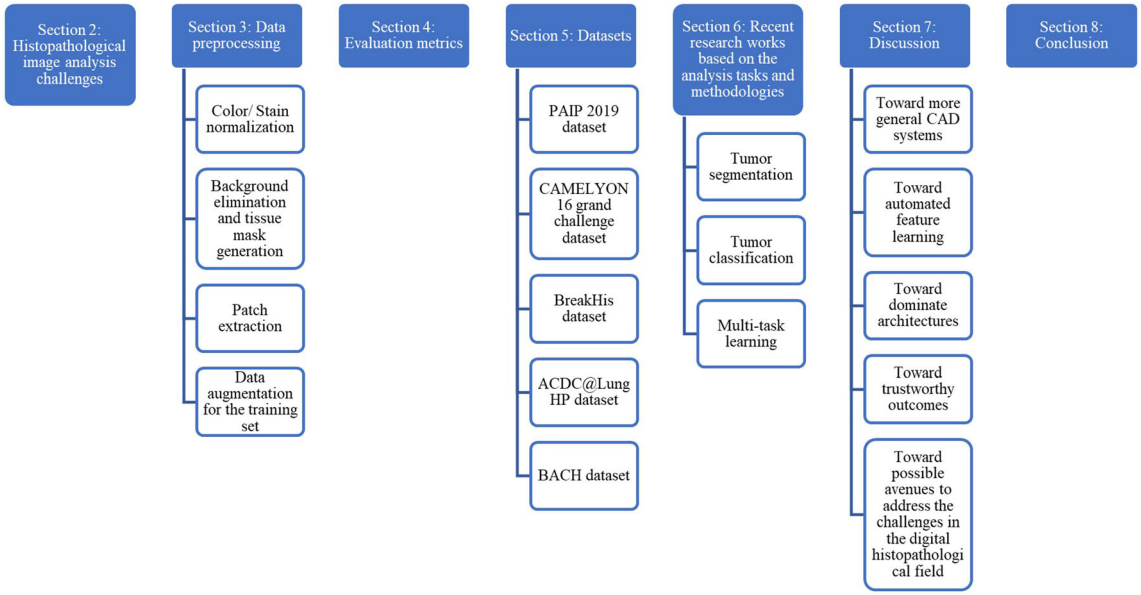


Fig. 2 The structure of the paper

2 Histopathological image analysis challenges

WSIs are considered among the most complex image modalities. A myriad of limitations and pitfalls bind the automatic analysis processes of WSIs and make them very challenging compared with the analysis of other traditional image modalities like CT or MRI. A discussion of some of these hurdles is provided next.

2.1 Large dimensionality of WSIs

The WSI size is enormous since it contains multiple resolutions. WSI is even sometimes called a giga-pixel scale image. Thus, even with the availability and the training acceleration provided by GPUs; GPU-based models can obtain up to 40 × time improvement over CPU-based models [25], the WSI analysis faces an immeasurable computational cost problem. In other words, loading the raw WSI dataset directly in the limited GPU memory is unfeasible since these slides are usually gigabytes in size. Even if part of the data is loaded successfully, the training process will be exhaustive and require many training days. Furthermore, many trials are also needed to test the different possible design choices of the tested model, such as the learning rate, the batch size, or the kernel sizes [25]. Thus, these slides pose a computation bottleneck in terms of storage and processing. Even more, since the size of the WSI is too big, they cannot be opened with regular image viewers even if they have a standard image format like (*.tiff*); they need special libraries such as OpenSlide[1] to process and view them.

2.2 Compatibility with deep learning algorithms

Deep learning models, especially convolutional-based models, are designed to receive a small fixed-size input (image). The WSI at the highest resolution, containing the cellular level details, is far too big to be fed directly. For example, the input size of the popular VGG-16, ResNet, and DenseNet networks is (224 × 224), and for Inception-v3, it is (299 × 299), while, for example, the average dimension of WSIs in the PAIP2019 dataset [26] is (53,860 × 42,900). One way to tackle this challenge is to feed the model with a lower-resolution version of WSI. However, [27] most of the needed critical local cellular-level features and details are lost [28], resulting in poor performance. A possible solution to address this challenge is built based on the divide and conquer principle with a sliding window strategy, in which smaller patches that hold a partial scope of the WSI at the highest resolution are sampled and fed to the segmentation model. The predicted patches are stitched back to form the WSI again at the inference stage.

2.3 Data imbalance

The tumor usually occupies a small fragment in WSI. Thus, data imbalance occurs when comparing the background,

[1] https://openslide.org/, Access Date: November 1, 2021.

foreground, and normal-tissue areas with the tumor in the image.

2.4 Scarcity of pixel-level annotated datasets

There is a lack in the available public high-quality pixel-level annotated histopathological datasets required for supervised learning methods usually used in CAD segmentation algorithms. The annotations for these histopathological images are very difficult to obtain; it is a laborious, time and effort-consuming task due to the gigantic size of the WSIs. Also, they need to be annotated by highly trained professional medical experts with good knowledge and experience. This limited amount of offered data increases the risk of model overfitting.

2.5 Acquisition issues

Image acquisition involves the process of tissue sampling, slide staining, and digitizing. As with any manual work prone to human error, the acquisition process may be affected by multiple aspects [28] that drastically change the tissue appearance by introducing unwanted anomalies that cause significant variations within the same sampled tissue. Consequently, this may hinder the performance of WSI analysis. Such factors include tissue preparation procedures such as the cutting or sectioning process [29] and staining process issues such as tissue thickness, stain concentration, fixation characteristics, and inconsistent temperature during staining [30]. Moreover, different commercial whole slide scanners across laboratories have different workflows, different slide preparations [30], and different scanner configurations and image compressions. Thus, there are multiple file formats adopted by those different vendors, among them: "*.svs*" (Aperio scanner), "*.ndpi*" (Hamamatsu scanner), "*.tiff*" (Philips scanner). Furthermore, sometimes blurry, noisy regions with shadows are formed in the WSI due to loss of microscope focus during the scanning process [31]. Thus, developing a generalized framework that is robust against stain pattern variability is challenging.

2.6 Variations in appearance

The single WSI has a high heterogeneity in its morphological structures and textures, resulting in considerable tissue features' variations. Reasons for these variations include tumor type, location, staging [32] and tissue preparation process inconsistencies. This increases the difficulty of the decision-making process. In Figs. 3 and 4, examples of different normal and tumor patches with different structures within the same WSI, as can be noticed, even the normal tissues have varying patterns.

2.7 Optimal resolution selection

The WSI is scanned using a digital scanner at multiple magnifications. All these magnification levels are stored in one file per image in a pyramidal structure where each level in this pyramid represents the same content but with a specific resolution. The term resolution in whole slide imaging is defined using the objective used to scan the slide and is measured in micrometers per pixel [33]. For instance, digital whole slide scanners generate slides with $\times$ 20 or $\times$ 40 magnifications (objectives) with resolutions of 0.5 mm or 0.25 mm per pixel, respectively, with a 24-bit color depth. This raises a tradeoff between the view scope and the resolution level. As the magnification level becomes lower, a bigger part of the image can be displayed on the screen. The image thumbnail is where an overview of the entire WSI fits on the screen, but the fine details are missing. As the resolution increases, a narrower part of WSI is displayed but with reasonable details; see Fig. 5 for examples of $\times$ 5, $\times$ 10, and $\times$ 20 magnifications. At the maximum magnification, the scope of the

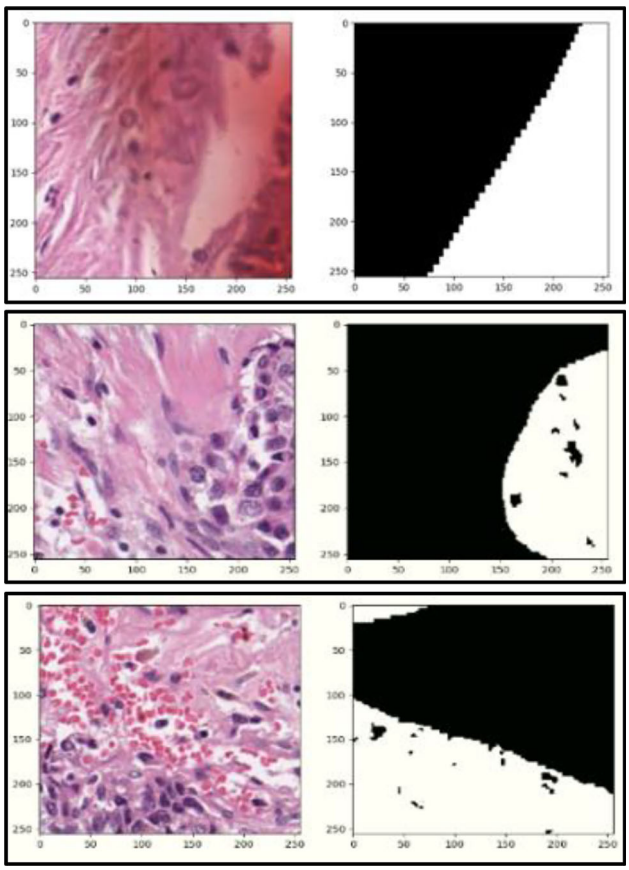


Fig. 3 Large variances in tumor patches of size 256 $\times$ 256 of an example WSI taken from the PAIP2019 dataset [26]. The right part represents the tissue image, while the left part represents the corresponding binary masks of these patches

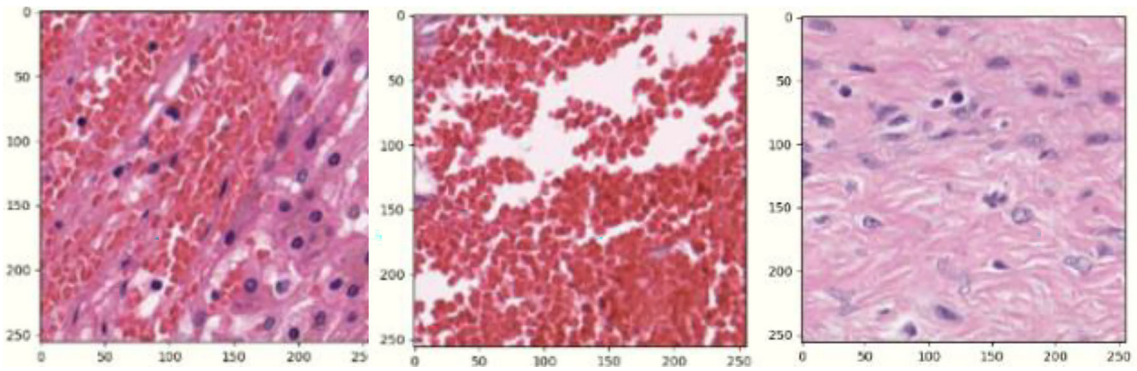


Fig. 4 Large variances in normal patches of size 256 × 256 of an example WSI taken from the PAIP2019 dataset [26]

image is very limited, but high-quality details of the cellular-level structure are obtained where the texture of the nucleus and the cytoplasm can be observed. Another tradeoff exists between the patch context (details level) and the resolution level. As the magnification level becomes lower, the patch contextual information becomes stronger. Thus, at the highest magnification, the contextual information is ignored, and it is hard to judge the tissue type of the patch. Therefore, selecting the best and most informative resolution level that suits the available computational requirements and the task at hand is difficult but contributes to the model's performance.

3 Data preprocessing

The preprocessing phase has the most impact on the overall segmentation results' accuracy and performance. In this phase, the raw data in the training set are prepared to be fed to the designed model. Preprocessing aims to increase homogeneity in the image, which helps accurately detect the regions of interest that contain the tumor by removing the unwanted obstacles. Also, this stage impacts the acceleration of the training and improves the generalization capabilities of the model.

3.1 Color/stain normalization

An essential step in tissue analysis is staining, differentiating the different colorless tissue components that reveal important information. The Hematoxylin and Eosin (H&E) stain is amongst the most common and popular staining techniques in histopathological slides due to its relative simplicity and ability to highlight and distinguish a wide range of tissue structures [34]. Hematoxylin selectively binds with nucleic DNA and RNA acids and dyes dense nucleic structures as dark blue or violet, whereas Eosin dyes cytoplasmic substances as pink. Other structures, including proteins, muscle tissues, and connective tissue fibers like collagen, are stained in varying shades, hues, and intensities of pink, orange, and red [34]. In addition to the wide range of variations of the stain colors due to the acquisition issues, the surrounding glass in the WSI background provides high brightness and may affect the mean and deviation of the WSI [30]. Consequently, this may mislead the segmentation result. Thus, color normalization is usually performed to homogenize the slide's visual properties; ensure that WSI follows a uniform scale; and compensate for the various staining conditions and slide thicknesses.

The staining techniques range from simple nonlinear pixel-value mapping operations to very complex models. A color space conversion from RGB to LUV spaces with color gradient flow estimation to handle the variations in staining was proposed in [35]. A three-phase New Complete Color Normalization (NCCN) method was proposed in [36], in which the first phase was the enhanced fuzzy illumination normalization to control the color enhancement and source-target image contrast difference after converting the RGB image to the LAB space. In the second phase, fuzzy-based stain normalization was performed. Finally, modified spectrum normalization was done in the third phase to preserve the histological structural information by calculating the stain concentration estimation in image channels. A template matching using an unsupervised sparse autoencoder, called StaNoSA, with the goal of clustering similar tissue types was proposed in [37]. Then, using the learned filters and histogram specifications, the target images were altered to match the source images.

Moreover, Generative Adversarial Networks (GANs) [38] were also utilized in stain normalization, such as the use of cycle-GAN in [39] for stain transformation that theoretically was able to generate changes in texture; also, the use of cycle-GAN in [40] that operated in stain-independent normalized color space. The Stain Acclimation Generative Adversarial Network (SA-GAN) proposed in [41] tackled the normalization process as an image-to-image translation task. SA-GAN consisted of one generator

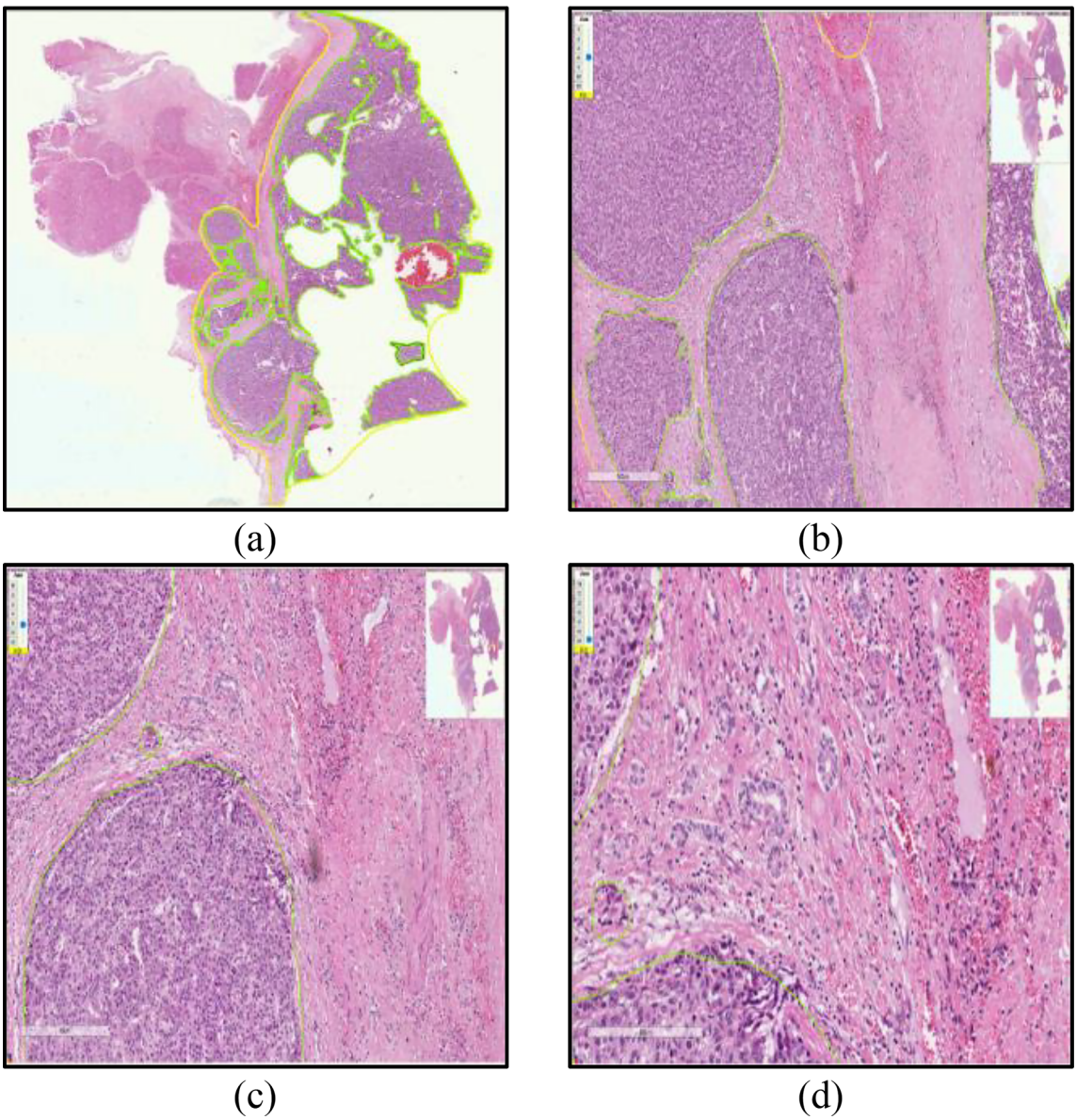


Fig. 5 An example WSI at different resolutions. **a** WSI image thumbnail: lowest resolution; **b** at magnification × 5; **c** at magnification × 10; **d** at magnification × 20: highest resolution. This WSI is taken from the PAIP2019 dataset [26]

and two task-specific discriminators, one to utilize the target image color statistics and patterns and the other for preserving the whole dataset distributions and structures. Thus, joint training of the input images and a novel designed color attribute metric, computed based on a reference image, was carried out by SA-GAN. On the other hand, re-staining a de-stained version of the stained image to provide normalization using a U-Net-based self-supervised learning model called RestainNet was introduced in [42]. The Hematoxylin and Eosin stains were extracted separately using Beer-Lambert's Law and then used to re-stain the L channel from the grayscale image counterpart that was converted to LAB color space. RestainNet used GAN loss to verify the quality and the correctness of the generated re-stained images. Fuethermore, StainNet framework with distillation learning was proposed in [43] for learning source-target image color mapping and relationships in the whole dataset. At first, StainGAN, the teacher network consisting of two generators and two discriminators, was trained to normalize the source images that were then fed as ground truth labels to StainNet, the fully convolutional-based student network. More details and methods of different stain normalization methods in the histopathological analysis field can be found in [44].

3.2 Background elimination and tissue mask generation

The WSIs are prepared by placing the biopsy sample on a glass slide; therefore, the large areas of WSI are irrelevant backgrounds that consume unnecessary computational costs without any added benefits. Several techniques are used for background removal and tissue mask generation, such as Otsu's adaptive threshold method [45] which performs clustering and optimizes the intra-class variance to separate the image pixels into background and foreground.

3.3 Patch extraction

One of the most used approaches for WSIs analysis is patch-based. There are many reasons for the popularity of these approaches; among them is the tremendous size of WSI that is incompatible with most of the deep models' specifications, reducing the memory requirements and the computational costs of dealing with the raw WSI, and solving the class imbalance issues by sampling equal patches from each class. Many patch sampling techniques exist in the literature, such as regular sampling, uniform sampling, or any other customized sampling strategy, such as the sampling done in [46, 47], or [48]. These techniques feed the analysis approach with the most appropriate patches necessary to boost its performance.

3.4 Data augmentation for the training set

Data augmentation techniques aim to address the annotation scarcity issue by obtaining new images that preserve the same semantic content of the original images without any loss in their distinctive information. These techniques solve the overfitting problem caused by the small training datasets by virtually enlarging their sizes [49]. Other appealing advantages of augmentation are partially alleviating the class imbalance issue and providing the model with the required robustness against any variations caused by the staining and the slide-making processes. The traditional ways for data augmentation are based on spatial and intensity information, such as the visual transformation by adding Gaussian noise, applying random brightness, random contrast, random gamma or translation; nondestructive transformations such as scaling, horizontal flipping, rotation; or non-rigid augmentation such as Elastic Transformation [50] where each pixel moves around independently based on distortion fields.

Recently, GAN also has become a popular method for data augmentation by generating synthetic images. However, the GAN-based augmentation models put extra effort into the available computation resources. Examples of their use in the histopathological analysis field include the pathologyGAN model [51] for artificial image generation from a structured latent space; the cycleGAN model [52] for normal to abnormal tissue translation; and the Progressive Growing of GAN (PG-GAN) model [53] for synthetic image patch generation. However, since the main aim in the generative-based models is to mimic, not learn, the original data distribution, the incorrect use of these generated images may have an inverse impact on the model performance.

4 Evaluation metrics

The Quantitative assessment of the validity and effectiveness of the tumor analysis algorithm in WSIs is evaluated using performance metrics. However, there is a lack of performance standardization in histopathological analysis research; in other words, the use of performance metrics is inconsistent. Different datasets and tasks use different measures. The most used metrics in this research field belong to the area error metric category, which evaluates the similarity between the tumor region generated by the designed algorithm and the actual tumor region that pathologists manually delineate.

Some standard metrics are widely used to assist the histopathological analysis model despite the carried analysis task, such as accuracy, precision, sensitivity, F1-score, and specificity; their mathematical formulas are enlisted in (1) to (5), where TP, FP, FN, and TN indicate True Positive, False Positive, False Negative and True Negative, respectively. However, some metrics are especially associated with a particular analysis task. For example, since segmentation models aim to draw precise tumor boundaries, the Dice score coefficient (6) and Jaccard index (aka Similarity Index or Intersection Over Union (IOU) (7)) are usually adopted for their evaluation in some public datasets. The classification models use the Receiver Operating Characteristic (ROC) and Area Under ROC (AUC-ROC) curves. On the other hand, a variant of the ROC curve called Free-response ROC (FROC) is used to identify multiple abnormal regions in the detection models. FROC replaced the original False Positive Rate (FPR) in the ROC curve with FP on its horizontal axis.

$$Accuracy\,(ACC) = \frac{TP + TN}{TP + FP + FN + TN}, \tag{1}$$

$$\text{Precision}\,(\text{Pr}) = \frac{TP}{TP + FP}, \tag{2}$$

$$\text{Sensitivity}\,(SEN) = \frac{TP}{TP + FN}, \tag{3}$$

$$F1\ \ score = \frac{2 \times (Pr \times SEN)}{Pr + SEN}, \tag{4}$$

$$Specificity\,(SPE) = \frac{TN}{TN + FP}, \tag{5}$$

$$Dice\ Score\ Coefficient\,(DSC) = \frac{2 \times TP}{2 \times TP + FN + FP}, \tag{6}$$

$$Jaccard\ Index(JI) = \frac{|Ag \cap Ap|}{|Ag \cup Ap|}, \tag{7}$$

where Ag is the ground truth tumor area determined by the pathologist, and Ap is the predicted area by the segmentation algorithm.

5 Datasets

One of the significant contributors to any deep learning-based system's success and good performance is the existence of large, high-quality, and well-annotated datasets to train and evaluate the proposed model. Next, some famous and widely used public datasets in the histopathological domain are described. Table 2 presents the dataset's characteristics, including the organ of interest, the data collection period, the annotation type, the statistics of training and testing subsets, the devices and the medical centers used to obtain the data, and evaluation metrics officially adopted by each dataset. Furthermore, the top performance achieved by the reviewed works in this survey in each dataset is also reported.

5.1 PAIP 2019 dataset

The PAIP (**P**athology **A**rtificial **I**ntelligence **P**latform) 2019 dataset [26], released as part of the MICCAI 2019 Grand Challenge for Pathology, consists of WSIs of diagnosed liver hepatocellular carcinoma. PAIP2019 aims for liver cancer segmentation and viable tumor burden estimation. The primary source of WSIs is the tissues taken from resection samples, except 9 WSIs were taken from biopsies. Each image identified two tumor areas: viable and whole tumors. The viable tumor areas correspond to viable tumor cell nests without the surrounding stroma. On the other side, the whole tumor is the outermost boundary that encloses all the viable tumor cell nests, along with tumor necrosis and tumor capsule. A total of 100 liver WSIs that H&E stains are randomly divided into training, validation, and test subsets with 50, 10, and 40 WSIs, respectively. The training and validation data have a pixel-level annotation for the viable and whole tumor regions. However, the test set annotations are unavailable; its results are evaluated through online submission on the challenge's website using the clipped Jaccard index metric that penalizes the scores of less than 0.65 by zeroing them out [26].

5.2 CAMELYON16 grand challenge dataset

The most commonly used public dataset for breast WSIs is the Camelyon16 (CAncer MEtastases in LYmph Nodes) Grand Challenge dataset [54], which was released as part of the International Symposium on Biomedical Imaging (ISBI). Camelyon16 aims to detect and classify metastatic breast cancer in stained WSIs of sentinel lymph node biopsies. The main source of Camelyon16 WSIs is the sentinel lymph node biopsies collected independently from two Dutch medical sources. A total of 399 breast WSIs, stained by H&E, are used in the Camelyon16 dataset, divided into 270 and 129 WSIs for training and testing, respectively. The training set contains 160 normal healthy slides and 110 slides that exhibited metastases. The training data have both pixel-level annotation, i.e., segmentation mask, and image-wise classification labels. However, the testing data are unlabeled, and the challenge organizers hold their annotations for performance evaluation according to two evaluation strategies: Slide-based and Lesion-based. The former uses AUC-ROC, while the latter uses the average FROC metric that detects the average slide level sensitivity at false-positive rates of 0.25, 0.5, 1, 2, 4, and 8.

5.3 BreakHis dataset

BreakHis dataset [55] consists of microscopic histopathological breast slides taken from the expelled breast tissues after surgical biopsies during a clinical study. BreaKHis dataset consists of 7,909 histopathological breast slides collected from 82 patients divided into 2,480 microscopic images with benign and 5,429 with malignant tumors. After that, each malignancy type is subdivided into four sub-categories corresponding to four different appearances of the tumor at the microscopic level, i.e., the four types of the benign tumor are Phyllodes Tumor, Tubular Adenoma, Fibroadenomas, and Adenosis. The four types of malignant tumors included in the dataset are Papillary Carcinoma, Mucinous Carcinoma, Lobular Carcinoma, and Ductal Carcinoma.

5.4 ACDC@LungHP dataset

The ACDC@LungHP (Automatic Cancer Detection and Classification in Whole-slide Lung Histopathology) dataset [64], which was released as part of the challenge at the IEEE International Symposium on Biomedical Imaging (ISBI) conference 2019, consists of H&E-stained WSIs

Table 2 Comparison of the details of the most widely-used public histopathological datasets

Name + year		CAMELYON16[a] (2016)	BreakHis[b] (2016)	BACH[c] (2018)	PAIP2019[d] (2019)	ACDC@ LungHP[e] (2019)
Organ		Breast	Breast	Breast	Liver	Lung
Task		Detection of metastases	Classification	Classification and Localization	Segmentation	Segmentation
Data collection period		The first half of 2015	January 2014 to December 2014	Microscopy: 2014, 2015, and 2017, WSI: 2013–2015	2005 to June 2018	January 2016 to November 2017
Annotation type		Pixel-wise	Image labels	Microscopy: image-wise labels (.csv file) WSI: pixel-wise label (only 10 WSI in.xml)	Pixel-wise (training: binary mask, validation:.xml format)	Pixel-wise (.xml file)
Source	Medical center	Radboud University Medical Center, Nijmegen, the Netherland, // University of Utrecht Medical Centre, Utrecht, the Netherland	Prognostics and Diagnostics (P&D) Laboratory, Pathological Anatomy and Cytopathology, Parana, Brazil	Microscopy: Hospital CUF Porto, Centro Hospitalar do Tâmega e Sousa and Centro Hospitalar Cova da Beira, Porto and Castelo Branco, Portugal., WSI: Centro Hospitalar Cova da Beira, Castelo Branco, Portugal	Seoul National University Hospital (SNUH), SNUBH, SMG-SNU BMC Korean medical centers	Department of Pulmonary Oncology in the First Hospital of Changsha, China
	Device	Pannoramic 250 Flash II scanner from 3DHISTECH// NanoZoomer-XR scanner from Hamamatsu Photonics	Olympus BX-50 system microscope with a relay lens With Samsung digital color camera SCC-131AN	Microscopy: Leica DM 2000 LED microscope and a Leica ICC50 HD camera WSI: Leica SCN400 scanner	Aperio AT2 scanner	Pannoramic 250 scanner from 3DHISTECH
	WSI format	.TIF format	.PNG format with no compression	Microscopy:.TIFF format WSI:.SVS format	.SVS format	.TIFF format
	Magnification	20x, 40x	400 × visual magnification/ 40 × objective lens	Microscopy: 400x WSI: 40x	20x	20x
Image distribution	#Training images	270	70% of total images	Microscopy:240 WSI: 30	50	150
	#Testing images	129	30% of total images	Microscopy: 80 (validation) + 80 (testing) WSI: 10	40 testing and 10 validation	50
	Total #images	399	9109 microscopic images (from website) // 7909 (from paper) with size 700 × 460	Microscopy: 400 with size 2048 × 1536 WSI: 40	100 WSI	200 WSI

Table 2 (continued)

Name + year	CAMELYON16[a] (2016)	BreakHis[b] (2016)	BACH[c] (2018)	PAIP2019[d] (2019)	ACDC@LungHP[e] (2019)
Official evaluation metric	AUC-ROC, FROC	ROC, Patient Recognition Rate	Custom evaluation score, Accuracy, Sensitivity, Specificity	Averaged Clipped Jaccard Index	Dice coefficient
Top achieved performance from reviewed papers in this survey	AUC-ROC of 0.988 and FROC of 0.902 achieved by [56], FROC of 0.86 achieved by [57], AUC-ROC of 0.9875 and FROC of 0.8533 achieved by [58]	Patient Recognition Rate of 97.5 achieved by [59]	Accuracy of 0.9375 achieved by [60], Accuracy of 0.9357 and sensitivity of 0.9426 achieved by [61]	Averaged Clipped Jaccard Index on testing set of 0.797 achieved by [62], and averaged Jaccard index on validation set of 0.7964 achieved by [30]	Dice coefficient of 0.797 achieved by [63]

[a]Download link: https://camelyon16.grand-challenge.org/, Access Date: November 1, 2021

[b]Download link: https://web.inf.ufpr.br/vri/databases/breast-cancer-histopathological-database-breakhis/, Access Date: November 1, 2021

[c]Download link: https://iciar2018-challenge.grand-challenge.org/, Access Date: November 1, 2021

[d]Download link: https://paip2019.grand-challenge.org/, Access Date: November 1, 2021

[e]Download link: https://acdc-lunghp.grand-challenge.org/, Access Date: November 1, 2021

with lung cancer. ACDC@LungHP aims for lung cancer detection and classification in two stages; the first stage focuses on the presence or not of lung cancer in the WSI, while the second stage focuses on classifying the main lung cancer subtypes: squamous cell carcinoma, small cell carcinoma, and adenocarcinoma. Since the detection in ACDC@LungHP is required on the pixel level, ACDC@LungHP eventually performs cancer tissue segmentation. 200 lung WSIs were stained by H&E and randomly divided into training and testing subsets with 150 and 50 WSIs, respectively. The training dataset has available pixel-level annotations, while the challenge organizers hold the testing images and annotations to perform unbiased performance evaluation using the Dice coefficient metric.

5.5 BACH dataset

The BACH (BreAst Cancer Histology) [65] dataset, which was released as part of a grand challenge by the 15th International Conference on Image Analysis and Recognition (ICIAR) 2018, consists of two types of breast histopathological images: microscopic and whole slide images. BACH aims for tumor classification and localization. The BACH dataset was divided into two parts and was extracted from different patients; part A is the H&E stained microscopic images extracted from patient ids 1 to 10, and part B is the WSIs extracted from patient ids 11 to 39. Part A of the BACH dataset consists of 400 image-wise annotated microscopic images subdivided equally into four classes: Normal, Benign, In situ carcinoma, and Invasive carcinoma. Each class in this part of the BACH dataset is randomly divided into 60, 20, and 20 images for training, validation, and testing sets, respectively. On the other hand, part B of the BACH dataset consists of 10 pixel-wise annotated and 20 non-annotated WSIs for training purposes, with 10 WSIs kept for testing.

6 Recent research works based on the analysis tasks and methodologies

Medical image analysis can be split into major overlapping tasks that can all be performed by a single CAD system, or each can be the target of a specific CAD system. This survey focuses on the segmentation, the classification and the muti-learning tasks. Moreover, each task is accomplished by various learning methodologies [66], such as Unsupervised learning, Weakly- and Semi-supervised learning, Supervised learning, and Reinforcement learning. The algorithms in unsupervised learning learn the desired output from the data directly without any predefined assumptions or labels by identifying the underlying data patterns. No correlation exists between the model's inputs and outputs [66]. In weakly-supervised learning, the algorithms learn from the coarse-grained labeled data, namely, data with image-level labels. Multiple-Instance Learning (MIL) is a widely used paradigm of weakly supervised learning methodology in which the image instances (patches) are combined in bags, with each assigned a single positive or negative label (coarse-grained annotation) that

describes its contents. While in semi-supervised learning, the algorithms benefit from incomplete or partially pixel-level labeled data with dependency on structural data assumptions to account for the unlabeled portion of the data, i.e., the algorithms apply the knowledge obtained from the labeled data on learning the unlabeled data [66]. Finally, in supervised learning, the algorithms are trained with fine-grained pixel-level annotated data that contain labels designating the desired output for each corresponding input. Finally, Reinforcement Learning embraces a teaching–learning technique trained to optimize the output by forcing rewards and penalties based on the algorithm performance [66]. Next, some of the recent notable works are discussed within each task based on their learning schemes and methodologies (single, hybrid, or ensemble methodology). Since there is no unified structure of these different works, then the comparison between them is difficult. Therefore, an overall description of their architectures is provided, followed by tables that provide a comparison of these methods' details according to the publication year, the organ, the adopted preprocessing and augmentation techniques, the used dataset, and evaluation metrics.

6.1 Tumor segmentation

Segmentation is the task of partitioning an image into non-overlapping distinguished objects and separating them from the image background. Usually, it aims to extract tumor regions and delimit their boundaries as precisely as possible in the medical images. The medical image quality directly and substantially affects the segmentation output. Following are summarizations of the recent works in this task categorized according to their methodologies into weakly- and semi-supervised learning-based approaches, reinforcement learning-based approaches, and supervised learning approaches. The latter is divided into four categories: transfer learning-based approaches, convolutional-based approaches, approaches that perform hybridization between the deep learning model with superpixel segmentation, with hand-crafted features, and with probabilistic models, and finally, ensemble approaches. Figure 6 illustrates this categorization with examples of works reviewed in this survey that belong to each category. While Tables 3, to 4, 5, 6 include more details and comparisons between them according to their characteristics.

6.1.1 Weak-/semi-supervised learning-based approaches

A weakly supervised learning segmentation framework called CAMEL was proposed in [67]. CAMEL had two steps: a label enrichment step based on combined MIL (cMIL) and an end-to-end Fully Convolutional Network (FCN) based segmentation step. CAMEL generated approximate pixel-level labels needed for supervised learning by automatically creating instance-level labels of the image's extracted latticed instances using cMIL. cMIL consisted of two cascaded MIL-based classifiers with different instance selection criteria (Max-Max and Max–Min) that operated on image-level labels. Afterward, those predicted instance labels, cancerous and non-cancerous labels, were given to the corresponding pixels within each instance by generating a pixel-wise heatmap per each patch. This step enriched supervision information in CAMEL by transforming the weakly supervised segmentation problem into a fully supervised classification task. Nevertheless, an over-labeling issue was reported since the predicted instance-level label was progressively assigned to the instance's pixels, causing ambiguous segmentation. Therefore, boundary refinement techniques may become necessary to enhance performance.

A semi-supervised expectation maximization-based segmentation approach constrained by the estimated prior distribution was developed in [68]. It initially trained the classifier with full supervision on a limited dataset with pixel-wise annotations to obtain a solid initialization of the model parameters. Then it utilized the only available weak annotations of another large-scale dataset, i.e., image-level labels, in this classifier to finetune it to leverage its valuable representations. By joining the strongly annotated limited-size dataset and the weakly annotated large-scale dataset, the semi-supervised learning (the second dataset) improved by incorporating prior knowledge as a bias (the first dataset). This semi-supervised model achieved a mean Jaccard index of 0.495, which outperformed the supervised model

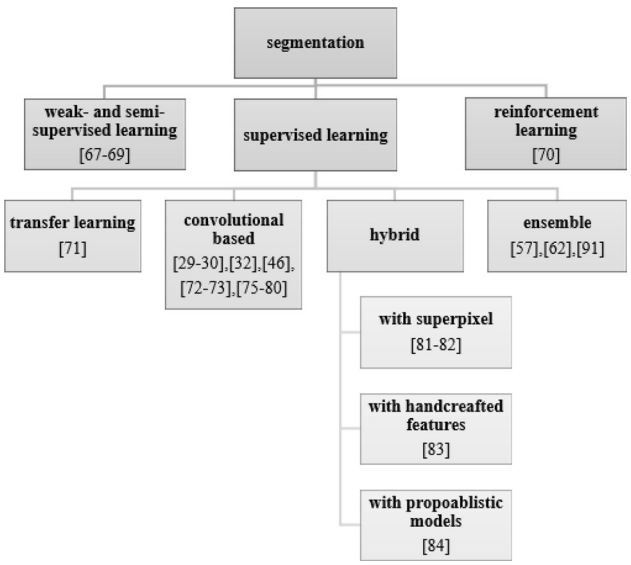


Fig. 6 Categorizing the recent works in the segmentation task according to the adopted methodology

Table 3 The details of the recent WSI tumor segmentation Semi-Supervised (SS), Weak Supervised (WS), and Reinforcement Learning (RL) based approaches

Nature	References	Year	Organ	Method	Preprocessing	Data augmentation	Dataset	Evaluation metric
WS	[67]	2019	Colon, Breast	CAMEL: MIL-based label enrichment framework	Patch extraction of size 1280^2 at 20 × magnification split into 320^2 and 160^2 patches	Rotation, mirroring, random scaling (between 1.0 × and 1.2x)	CAMELYON16 and a private colorectal adenoma dataset consisting of 177 WSIs	CAMELYON16 with (U-Net): Sensitivity (0.947), Specificity (0.941), Accuracy (0.943), F1-score (0.918), IoU (0.848)
SS	[68]	2018	Prostate	Expectation–Maximization based semi-supervised deep learning	Color normalization using transfer algorithm	–	Private collection of 135 fully annotated and 1,800 weakly annotated tiles	Mean Jaccard Index (0.4947), overall Pixel Accuracy (0.7479)
Hybrid WS and SS	[69]	2019	Basal Cells (skin)	Weakly- and semi-supervised Graph-CNN (GCN)	Divide each image into varying patch sizes of 11 × 13 to 12 × 16	–	A private dataset consists of 1241 Basal Cell Carcinoma images	ROC-AUC (0.9538), mAP (0.9583), Accuracy (0.9027), Precision (0.9037), Sensitivity (0.8848), Specificity (0.9182)
RL	[70]	2018	Breast	RAZN: Reinforced Auto-Zoom Net	Patch extraction of size 256^2 at two resolutions	–	BACH 2018 dataset/ WSIs	Mean Intersection Over Union (mIOU) (0.49)

by 14%. However, the model was built based on the assumption that the tiles within the contour hold the same image level.

A combination of weakly- and semi-supervised learning was used to develop a graph convolutional neural network [69] to address the issue of interpretable Basal Cell Carcinoma segmentation in pathological images. The mixture of the two supervision types: weak and semi, provided flexibility by utilizing the only available image-level labels as in the former and benefiting from the limited number of images with pixel-wise segmentation labels as in the latter. The proposed framework consists of two modules; the first module was a convolutional graph module with nodes representing the patch-level features extracted using pre-trained Resnet-50 model, and edges represented the topological relationship between them; it was used to propagate the supervisory information over patches to learn patch-aware interpretability by forming per patch probability score. On the other hand, the second module consisted of an aggregation function that obtained the image-level predictions by combining the prior expert knowledge with the structural information obtained in the patch-level predictions aggregation. The prior knowledge bridged the gap between the patch-level and the image-level predictions by providing directed information about the cancer cells' patterns and behaviors.

6.1.2 Reinforcement learning-based approach

To mimic the typical pathologists' zoom-in operation of the suspicious areas when examining WSI, Reinforced Auto-Zoom Net (RAZN) segmentation framework was introduced in [70]. RAZN attentively zooms in on regions of interest, i.e., tumorous areas, using reinforcement learning. Thus, to capture the different cancer representations in different fields of view, RAZN consisted of multiple policy networks and a segmentation network. There existed a network for each zoom-in level to tackle the scale-invariant limitation of CNNs where each network only deals with a specific magnification. A policy gradient network was learned to either apply a zoom-in action in the abnormal tissue and thus examined more local information or skip processing the normal and background tissues. This selectivity made RAZN more robust and increased the accuracy; it achieved an mIOU of 0.49 while fastening the inference time. However, the requirement of separating the policy networks for every zoom-in level resulted in increasing the number of trainable parameters in the overall architecture and, at the same time, prevented the interaction and information flow between the different resolutions. Moreover, in this selective processing, vast areas in WSI remain unexplored, which hinders the optimization if important information is missed.

Table 4 The details of the recent Convolutional-based WSI tumor segmentation approaches

References	Year	Organ	Method	Preprocessing	Data augmentation	Dataset	Evaluation metric
[72]	2018	Breast	MRN: multi-Resolution U-Net based Networks	Co-centric patch extraction at multiple levels	–	Camelyon16 Grand Challenge Dataset	ROC curves
[46]	2018	Breast	Patch-based CNN with High-throughput Adaptive Sampling for Histopathology Image (HASHI)	Color transformation and normalization, sampling of patches with sizes of 101^2 from tissue regions only	–	Three Public datasets with 500 cases and 195 additional cases from The Cancer Genome Atlas (TCGA[a])	Dice coefficient (0.76), Positive Predictive Value (PPV) (0.72), Negative Predictive Value (NPV) (0.97)
[73]	2019	Breast, prostate	Dense multi-path decoder	Patch extraction	Random rotation, shear, shifting	Breast and prostate tissue datasets: 108 and 52 H&E breast tissues images and 224 H&E prostate tissue images	Pixel-wise accuracy (breast: 0.9122, prostate: 0.9074), Rand Index score (breast: 0.8398, prostate: 0.8320), AUC (breast: 0.9716, prostate: 0.9719), Dice coefficient for stroma (breast: 0.9092, prostate: 0.9021), Dice coefficient for epithelium (breast: 0.9150, prostate: 0.9121)
[29]	2019	Kidney	CNN cascades	Data normalization to zero-mean and unit-variance	Rotation, flipping, elastic deformations	24 WSIs of resected mouse kidneys stained with PAS	results of best performing cascade: Dice similarity coefficient (0.90), Precision (0.89), Recall (0.91)
[32]	2020	Breast	DCNN: deep convolutional neural network along with separable atrous convolution	–	Random zooming, cropping, and flipping	Private dataset of 12 breast H&E WSI at 40 × magnification level with a total of 640 patches of 500 × 500 pixels	Frequency Weighted Intersection over Union (FWIoU) (0.9252), Accuracy (0.9562), mIoU (0.6452), Sensitivity (0.8858), Specificity (0.9739)
[75]	2021	Breast	DMMN: Deep Multi-Magnification Network	–	Elastic deformation, rotation, vertical and horizontal flipping, brightness, contrast, color jittering	Two Datasets: Triple-Negative Breast Cancer (TNBC) contains 38 images; Dataset-II: 10 WSIs from lumpectomy and breast margins	mIOU (Dataset-I: 0.870, Dataset-II: 0.706), mean recall (Dataset-I: 0.939, Dataset-II: 0.898), mean precision (Dataset-I: 0.922, Dataset-II: 0.795)
[76]	2019	Lung	AWMF-CNN: Adaptive-Weighting-Multi-Field-of-View-CNN	Patch extraction at three different magnifications (20x, 10x, 5x) of size 256^2 with the same region	Random horizontal and vertical flipping	Private collection of 29 WSIs of sliced lung adenocarcinoma	Overall Pixel Accuracy (2-class: 0.916, 4-class: 0.672), mean of per-class Accuracy (2-class: 0.904, 4-class: 0.676), mean IoU (2-class: 0.821, 4-class: 0.536)

Table 4 (continued)

References	Year	Organ	Method	Preprocessing	Data augmentation	Dataset	Evaluation metric
[77]	2020	Lung	CSC-NET: Cancer Sensitive Cascaded Networks	Patch extraction of two sizes: 512^2 and 1024^2, to train the two networks	–	Public ACDC@ LungHP ISBI2019 challenge dataset	Dice coefficient (0.694), Recall (0.792), Precision (0.947)
[78]	2021	Lung	HookNet: multi-branch encoder-decoder network	Concentric patch extraction at two different spatial resolutions	Spatial, color, noise, stain augmentations	Private breast dataset with 86 tissue sections; public (TCGA-LUSC) dataset with randomly selected 27 slides	F1-score (public dataset: 0.73, private dataset: 0.91)
[30]	2021	Liver	Multi-scale U-Net based approach with a voting strategy	Random patch cropping of size of 448^2, Partial color normalization, and weighted overlapping	Flipping, rotation, cropping, translation	MICCAI PAIP 2019 Challenge Dataset	Jaccard score (validation set: 0.7964)
[79]	2021	Liver, breast	Multi-encoder Fully-convolutional Networks with deep fusion	Color normalization, then random patches were redrawn after each epoch	Rotation, horizontal flipping, saturation, and brightness transformation	MICCAI PAIP 2019, Camelyon16, BACH 2018 Challenge Datasets	Fivefold cross-validation on training set: Averaged Jaccard Index (PAIP2019: 0.865, BACH: 0.579, Camelyon16: 0.889)
[80]	2021	Liver, colon	SAFS: Structure-aware Scale-Adaptive Feature Selection method	Patch extraction	Color jitter, random rotation, horizontal, vertical flipping	MICCAI PAIP 2019 and PAIP 2020[b] Challenge Datasets	Fivefold cross-validation on training set: Dice coefficient (PAIP2019: 0.9006, PAIP2020: 0.9246), Jaccard index (PAIP 2019: 0.8211, PAIP2020: 0.8614), clipped Jaccard index (PAIP2019: 0.7989, PAIP2020: 0.8422)

[a]The Cancer Genome Atlas (TCGA) repository contains high-volume genomic and clinical data with approximately 11 k cancerous tissue slides of multiple organs. TCGA is a joint project of the National Cancer Institute (NCI) and the National Human Genome Research Institute (NHGRI). TCGA provides only identifying labels of the cancer types in the WSI with no available cancerous tissue segmentation annotations. Download link: tcga-data.nci.nih.gov, Access date: November 1, 2021

[b]The PAIP 2020 dataset for colorectal cancer molecular subtypes classification consisted of 47 training WSIs with corresponding annotated binary scanned at 40 × magnification. PAIP2020 also contains 31 unannotated validation and 40 testing WSIs

Table 3 summarizes the different details of the various tumor segmentation methods that belong to the weakly- and semi-supervised and reinforcement learning categories.

6.1.3 Supervised learning-based approaches

6.1.3.1 Transfer learning-based approach A transfer learning-based step-wise approach was introduced in [71] to reduce false positives and increase accuracy. It consisted of two steps: the germinal centers' exclusion step and the tumor cell detection step, each performed using a separate classifier. A CNN-based classifier with a random forest algorithm was used in the first step to eliminate lymphoid follicles using two-class labels (tumor and non-tumor). Then a new class label was added to this approach containing the cause of the false positive. i.e., Lymphoid follicles that are often misclassified as noncancerous regions. Thus, three class labels were fed to a convolutional-based classifier that was pre-trained on the ImageNet dataset in the second step to detect cancer cells precisely. They reported a limitation of their used image analysis platform (HALO) to have a fixed 0.5 threshold when binarizing the

Table 5 The details of the recent Hybrid based WSI tumor segmentation approaches, where HSP (Hybrid with SuperPixel), HHF (Hybrid with Handcrafted Features), HRC (Hybrid Recurrent and Convolutional), and Hybrid with Probabilistic Models (HPM)

Nature	References	Year	Organ	Method	Preprocessing	Data augmentation	Dataset	Evaluation metric
HSP	[81]	2018	Kidney	Hybrid of Superpixel Segmentation Algorithm and Deep Learning Method	Noise removal using the median filter	–	810 Kidney carcinomas histopathological images with 40 × magnification selected from TCGA data portal	Confusion matrix
HSP	[82]	2019	Breast	Superpixel-based SVM feature learning classifier + CNN Segmentation Network	Tile extraction of size 512^2	–	127 WSIs of breast tumor patients gathered from Acıbadem hospital and Warwick competition	Accuracy (0.9482), Precision (0.9283), Recall (0.9922)
HHF	[83]	2019	Colon	Persistent Homology and Deep Convolutional Features	Patch extraction of size 256^2	Rotation, horizontal and vertical flipping, perturbing hue variation of the color distribution	Two private colorectal datasets with 40 × magnification: Warwick-UHCW with 75 WSIs; Warwick-Osaka Dataset with 50 H&E tissue	Average patch-level F1-score (UHCW: 0.9243, Osaka: 0.8273),Precision (UHCW: 0.9267, Osaka: 0.8311), Recall (UHCW: 0.9220, Osaka: 0.8235)
HPM	[84]	2018	Breast	NCRF: Neural Conditional Random Field framework	Background removal using Otsu algorithm, extract random patch of size 768^2	Color jitter, random flipping, random rotation	Camelyon16 Grand Challenge Dataset	FROC (0.8096 based on ResNet-18)

generated heatmap, which caused high sensitivity and low specificity.

6.1.3.2 Convolutional-based approaches Two Multi-Resolution Networks (MRN) were developed in [72] that used base U-Net architecture. MRN used concentric patches with different resolutions fed to multiple encoders and had two options for the single decoder upsampling; either used bilinear interpolation "MRN-bilinear" or transposed convolution "MRN-transposed". Their results showed that MRN-transposed had a superior performance than MRN-bilinear due to the parameterization nature of the transposed convolutions. The multiple feature maps aggregated at each depth in the model resulted in information redundancy that put more pressure on the computation resources, especially the memory. Moreover, since the different magnification's feature maps were center cropped, this limits the benefits obtained from the information at boundary regions.

A High-throughput Adaptive Sampling for whole-slide Histopathology Image analysis (HASHI) framework was developed in [46]. It involved integrating a designed adaptive sampling strategy built based on the probability gradient and quasi-Monte Carlo sampling and a CNN-based classifier for representation learning. The adaptive strategy provided a guided sampling focused on extracting tiles from high-uncertainty areas in the image. These ambiguous regions contained a heterogeneous mixture of morphological and architectural features that needed more attention. Initially, using pseudorandom sampling, the first set of tiles was extracted from WSI to be fed to a CNN-based classifier to generate a heat map highlighting the uncertainty regions in WSI. Then, another set of tiles was densely extracted from these uncertain regions using the designed adaptive sampling strategy to produce the final probability map. The overall framework achieved robust and reproducible predictions with an average Dice coefficient of 0.76.

An enhancement of the decoder was done in [73] in the epithelium and stroma segmentation network. The Dense Decoder (DD) aimed to strengthen the learning and inference capabilities using a multi-path layout with a split-

Table 6 The details of the recent ensemble and transfer learning (TL) based WSI segmentation approaches for different organs

Nature	References	Year	Organ	Method	Preprocessing	Data augmentation	Dataset	Evaluation metric
Ensemble	[57]	2021	Liver, breast, colon	Ensemble of: DenseNet-121, Inception-ResNet-V2, and DeeplabV3Plus	Tissue mask generation using Otsu's thresholding. Random patch extraction of size 256^2	Horizontal and vertical flipping, rotation, Gaussian blurring, color augmentation	Camelyon16, Camelyon17[a], DigestPath, PAIP2019	Dice score (DigestPath: 0.782),Cohen's kappa score (Camelyon17: 0.9090), Average Jaccard Index (PAIP2019: testing set 0.750), FROC: (Camelyon16: 0.86)
Ensemble	[62]	2021	Breast, colon, liver	Ensemble learning scheme with selective kernel modules and spatial and channel-wise squeeze-and-excitation modules	Remove non-tissue regions using Otsu method, non-overlap feature extraction of size 256^2	Horizontal and vertical flipping, random rescaling, random cropping, random rotation	PAIP 2019, CRAG [93] (Colorectal denocarcinoma Gland) dataset with 213 images, public HCMC& CWRU dataset with 110 breast images	Jaccard index (PAIP2019: testing set 0.797), Dice score (CRAG: 0.923, HCMC&CWRU: 0.765)
Ensemble	[91]	2022	Liver	Enhanced U-Net with cross attention	Remove non-tissue regions using Otsu threshold, random patch extraction of size 256^2	horizontal and vertical flipping, Gaussian blur, sigmoid contrast	MICCAI PAIP 2019 Challenge Dataset	Clipped Jaccard index (testing set 0.6274)
TL	[71]	2019	Lung	Two-step deep learning approach	Random patch extraction of size 435^2 with equal probability for each class	Random rotation and shifts in hue, saturation, contrast, brightness	Private 349 whole-slide lung cancer lymph node images	ROC curve using foci size thresholds

[a]Camelyon17 dataset invigorated the Camelyon16 dataset by moving the analysis from slide level to patient level analysis with a total of 1000 slides for both training and testing collected from five medical centers with five slides per patient. Camelyon17 is partially pixel-level annotated

transform-merge strategy to form a more structured feature representation. DD was also equipped with dense shortcut connections between its layers to integrate global and local information.

Two different CNN cascades to segment the glomeruli in Kidney WSIs were introduced in [29]. In each cascade, two networks were independently optimized; their architecture consisted of the original U-Net, a shallow U-Net, and a Sliding-window convolutional neural network (SW-CNN). These two networks differ in the distribution of their input patches; the first network sampled random patches conditioned to contain part of the glomerulus, while the second took the samples randomly from all the kidney. The idea behind the two cascades was to localize regions of interest in a low-resolution image and precisely segment these detected regions in a high-resolution image.

An encoder-decoder-based Deep Convolutional Neural Network (DCNN) with separable Atrous convolution was developed in [32] for breast cancer segmentation. DeeplabV3Plus model [74] was used as the encoder backbone. Using the Atrous convolutions gave the model the required flexibility through adaptive controlling of the field of view with different kernel rates at low costs. Two tiles merging approaches were tested; the first approach divided the image in the inference stage into non-overlapping tiles. The whole segmentation map was constructed based on the tile segmentation results; however, this introduced noise and discontinuity issues at boundaries. On the other hand, the second approach extracted 50%-overlapped tiles and

averaged the generated tile-wise probability maps in the overlapping areas. Also, a CRF was applied to ensure continuity and consistency at boundaries.

A Deep Multi-Magnification Network (DMMN) was presented in [75] for breast WSI tissue segmentation. DMMN had multi-encoder, multi-decoder, and multi-concatenation architecture with an expanded view field due to input patches with various resolutions. High magnification patches provided cellular level details, while the low ones exhibited the tissue types' distribution and the architectural tumor growth patterns, thus constructing richer features in the intermediate layers. DMMN was reported to be sensitive to the different slide backgrounds generated by different scanners, which questioned the model's generalizability.

An Adaptive-Weighting-Multi-Field-of-View-CNN (AWMF–CNN) was developed in [76] that tackled the "resolution vs. field of view size" trade-off issue by adaptively aggregating the contextual features of three different magnifications of the image. AWMF-CNN consisted of three expert CNNs, a weighting CNN, and an aggregating CNN. Each expert CNN was trained independently to optimize the segmentation of a certain magnification. Then these pre-trained expert networks were re-trained during an end-to-end learning pipeline. Since the importance of the information obtained at each magnification level differs according to the input image, a weighting CNN was used to quantify the weights of each expert CNN. Then, the generated heat maps of these experts and their estimated importance were combined using the aggregating CNN to produce the final segmentation result. However, the neural-based fusion method complicated their model and caused information redundancy.

Cancer Sensitive Cascaded Network (CSC-Net) was developed in [77]. It consisted of two U-Nets; the first U-Net aimed to reduce the computational cost of the CSC-Net by filtering the background areas and segmenting the cancerous regions. Contrarily, the second U-Net refined these cancerous segmented regions and removed the miss-segmented ones to reduce the false-positive rate. Moreover, to achieve a high recall of the first U-Net and a high accuracy of the second, they designed a specific Cancer Sensitive Loss (CSL). However, the model was not end-to-end.

The HookNet [78] used concentric patches extracted at multiple resolutions, and it was composed of two encoder-decoder branches trained end-to-end; the context and target branches. The contextual information was captured in the context branch through a wider field of view, utilizing low-resolution input patches. While the target branch targeted the fine-grained visual details using the small field-of-view offered by the high-resolution input patches. The two branches share the same architecture but not the same weights. They also designed a hooking mechanism to concatenate the intermediate representations from both branches to produce a final segmentation map that combined context and details. Also, skip connections were used for localization support. Since the two branches were fed with concentric patches, the obtained features must be spatially aligned, which may conflict with the operation of the max-pooling layers. Consequently, the lower-resolution features may need to be cropped, resulting in losing the required global contextual information.

A multi-scale approach for liver cancer segmentation in WSIs was developed in [30] to capture the texture information of WSI at its different resolutions. It constructed a seven-level gaussian pyramidal decomposition representation for each WSI. The higher-level pixels were formed using Gaussian weights and the contribution of the five lower-level pixels. The size of an image at a certain level is one-fourth its size at the previous level. This involved the construction of seven training datasets, one for each level, each used to fine-tune a pre-trained U-Net model. Then, the seven distinct segmentation maps were combined to consider the local and macro features at different scales when obtaining the final prediction using shifted cropping technique, weighted overlapping assembling method, and voting mechanism. However, these different levels of predictions, which were obtained independently, sometimes produced complementary results that may violate the strength of the final ensemble prediction.

A family of multi-scale multi-encoder fully-convolutional neural networks with deep fusion was introduced in [79]. They utilized a spatial relationship preserving a combination of local and global contextual features from different scales. A base multi-scale merging block and a context classification gate block were designed to fuse the multiple encoders with different spatial scales in a spatially constrained fashion and gate the segmentation output. Then, these blocks were injected in different positions in the baseline Res-U-Net architecture to construct this multi-scale FCN family. For instance, a network named msY-Net was developed with two encoders with patches of two different scales merged at the U-Net bottleneck to be inputted to the single decoder. In addition, to integrate the global (large scale) and local (small scale) context, the msY^2-Net network used a sequential cascade of the merging blocks. On the other hand, the msYI-Net network had an additional large-scale context encoder parallel to the msY-Net without any merging, which generated two paths: full resolution segmentation and classification that were joined in the context-classification gate. Despite the development of many models, there was no proof of the substantial benefits of the early or multiple merge connections that existed in some of them. Furthermore, a

customized classification loss was introduced to stabilize and speed up the training using deep guidance.

The structure-aware scale-adaptive feature selection method was developed in [80]. To obtain robust features in this encoder-decoder architecture that describe the non-rigid vague tumor boundaries, the scale-adaptive feature selection (SAFS) module produced a dynamic weighted contribution of multiple features at different scales decoded at various stages. Thus, to generate the final prediction, SAFS replaced the use of a single feature or the use of simple unweighted feature fusion. Moreover, a structural similarity (SSIM) loss was adopted to regulate the network training process to account for the segmentation of small regions and obtain enhanced tissue structure awareness. The structural similarity focused on minimizing the difference in predicting the tumor boundaries' pixels to capture the tumor structure. Finally, a novel selective-kernel (SK) convolution was utilized to substitute the basic convolutions with attention in the proposed architecture to achieve large adaptive receptive fields.

As shown in Table 4, the multi-resolution convolutional-based models dominate the reviewed tumor segmentation methods over a single resolution. This resulted from the aid offered by the global information integrated from multiple scales to obtain richer contextual information. The breast also is the most examined organ in this category. There are various methods to integrate multi-resolution in the segmentation approaches, among them: using different receptive field sizes to obtain the spatial pyramid representations, injecting concentric patches at different levels in the encoder or the skip connections, or having multiple encoders assigned for each resolution with one or multiple corresponding decoders.

6.1.3.3 Hybrid approaches

- Hybrid with superpixels

  The SLIC superpixel method was combined in [81] in a convolutional based histopathological image segmentation. Their framework consisted of two stages: pre-segmentation using the SLIC clustering method to generate cellular and non-cellular superpixels, then the convolutional-based classification of these generated superpixels to output the final WSI segmentation with an overall distribution accuracy of 98.76%. However, an extra analysis of the superpixels' optimal number and diameter was required.

  A framework to segment IHC stained breast cancer images was developed in [82] that performed a two-step process; epithelial and stromal regions in WSI were classified using SLIC superpixel-based support vector machine classifier, then segmentation of membrane regions in the classified epithelial regions was done using convolutional auto-encoder segmentor. Two handcrafted features were combined and used as discriminative representatives for each generated superpixel that were then fed to the supervised classifier; local color histogram and Local Binary Patterns (LBP) superpixel-level features.
- Hybrid with handcrafted features

  A hybrid framework that enhanced CNN with statistical topological handcrafted features for colorectal cancer segmentation was presented in [83]. More specifically, the Persistent Homology Profiles (PHPs) were computed for each image patch to distinguish tumor and non-tumor cells by modeling the distribution and the degree of nuclear connectivity and divergence. They introduced two variants of the segmentation method with conflicting goals. The first algorithm optimized the speed with reasonable accuracy; the PHPs for tumor and normal exemplar patches were computed and compared with the PHPs of the rest of the WSI patches using symmetrized Kullback–Leibler Divergence (KLD). A variant of the KNN classifier was used for patch label prediction. The selection of the exemplar patches was made using CNN activations. On the other hand, their second segmentation method optimized the accuracy by feeding the PHP features and a set of learned deep convolutional features through two separate Random Forest regression models, then using a multistage ensemble strategy to fuse the final feature vector. However, the necessity of manually selecting the k parameter value in the classifier is considered a limitation.
- Hybrid with probabilistic models

  A probabilistic graphical model called Neural Conditional Random Field (NCRF) was developed in [84]. NCRF combined two stacked components in a unified model; fully connected CRF on top of the CNN feature extractor. Therefore, NCRF considered the neighboring patches' spatial relationships to enhance the quality and smoothness of the generated probability map. At first, the input image was divided into a grid of patches and fed to ResNet-18 and ResNet-34 networks to encode each patch into a comprehensive feature vector representation. Then CRF took the resulting grid and extracted the spatial relationships between them. The backpropagation method was used in the end-to-end training of the NCRF model, which introduced an increase in computational complexity.

Table 5 details the characteristics of the reviewed prominent hybrid methods applied for tumor segmentation. The tumor segmentation approaches that combine deep learning with additional aid aim to retrain all the slide information to the greatest extent and capture them

according to different points of view. This was utilized by using hand-crafted features or clustering techniques like superpixels. Furthermore, a fusion of different deep learning models, such as convolutional and recurrent models, was introduced to enhance the performance.

6.1.4 Ensemble approaches

An ensemble of three encoder-decoder-based FCN architectures was developed in [57] for tumor segmentation. The ensemble model independently fine-tuned the following pre-trained networks on different subsets of the training data: U-Net with DenseNet-121 [85] encoder, U-Net with Inception-ResNet-V2 [86] encoder, and DeeplabV3Plus [74] with Xception [87] network as the encoder. The first two models were trained on the ImageNet dataset [88] and the third using PASCAL VOC dataset [89]. In the inference stage, the patch coordinate sampling grid concept was introduced and applied to the post-processed tissue mask of the WSI to generate overlapping larger patches (than used in training). The final tumor prediction map was generated by averaging and stitching the constructed posterior probability maps of the three models in the ensemble. This was used to address the loss of context information at patch borders. Moreover, they used a hybrid loss function to tackle the class imbalance issue by combining cross-entropy loss and Dice loss functions in a linear combination.

An ensemble of two complementary models based on Selective Kernel Modules (SKMs) and spatial and channel-wise Squeeze-and-Excitation Modules (scSEMs) was proposed in [62]. A multi-task and ensemble learning strategies were used to capture distinctive feature representations from different perspectives. Their encoder-decoder-based proposed framework had a shared encoder (SE-ResNeXt-101 model [90]) and a decoder with three task-specific branches: tumorous patches pixel-wise segmentation branch, tumorous and normal patches pixel-wise segmentation branch, and patch-level classification branch. These branches were jointly trained end-to-end but with different loss functions associated with each of them. They aimed to expand the feature space using the information obtained from the individual branches and then extracted the general shared features needed to reduce the overfitting risk. Afterward, feature aggregation was performed by leveraging the advantages of the ensemble learning of two models in the decoder part. The SKM-based model segmented the varying-size tumors through multi-scale and multi-context feature extraction by changing receptive field sizes. On the other hand, the scSEMs-based model captured two types of interdependencies among features; spatial-wise and channel-wise.

In order to exploit, aggregate, and exchange the local and global contextual information from different parts, a cross-attention mechanism was incorporated in an enhanced U-Net in [91]. The used cross attention coupled the nonlinearities obtained from two branches from different levels in U-Net in a cross fashion and produced a weighted combination. Moreover, dilated convolution with bottleneck structure was used to generate compact summarization vector of the encoder features in the center of U-Net that exploited different scales without increasing the model capacity. Then, an ensemble strategy based on separate slide color contributions was also developed that implied separating the slide's different color components and generating three new versions by duplicating each color across the three channels. Afterward, four independent models were trained on the resulting versions in addition to the original slide, and the final segmentation prediction was generated by averaging the four separate predictions.

The final prediction in the ensemble model is usually obtained by combining multiple trainable deep models to enhance the predictions' accuracy and improve the constituent models' robustness. Table 6 summarizes the different ensemble models proposed for tumor segmentation. Some used the same structure but with different parts of the data or different magnification levels, while others used different architectures with the same data. As can be observed from the table, these ensemble-based models examined and analyzed more than one organ, which indicates that these methods are robust and address the generalizability requirement. Furthermore, to circumvent the poor and ambiguous delineation of FCN, some tumor segmentation approaches employed post-processing techniques, such as CRF, to refine the boundaries of the model's prediction iteratively.

6.2 Tumor classification

Classification is the task of classifying the important features and information extracted from the Region of Interest (ROI) that best describe the tumor. The ROI can be the output of the segmentation task, or a pathologist can manually select it. These features are fed to a classifier to determine the diagnosis and tumor category, i.e., define an image-level describing label. The correctness of the classification depends on the precise and accurate ROI selection. Following are summarizations of the recent works in this task categorized according to their methodologies into unsupervised, weakly-supervised, reinforcement, and supervised learning-based approaches. The latter is also divided into six categories: transfer learning-based approaches, convolutional, recurrent, generative, and graph-based approaches, and approaches that perform hybridization with handcrafted features, with probabilistic models, and with traditional machine learning methods.

Figure 7 illustrates this categorization with examples of works reviewed in this survey that belong to each category. While Tables 7, to 8, 9 include more details and comparisons between them according to their characteristics.

6.2.1 Unsupervised learning-based approach

A two-stage classification method was proposed in [92]. The first stage consisted of an unsupervised vector embedding of WSI patches using the encoder. A feature graph was formed by considering these embeddings' relative spatial displacements to form a stack of high-level feature maps per WSI. Three encoder types were investigated: A Convolutional Automatic Encoder (CAE) that minimized the mean squared error, a Variational.

Automatic Encoder (VAE) that minimized the KL-divergence loss, and a new proposed encoder based on contrastive training with the construction of an artificial training dataset consisting of positive and negative pairs. These compact representations were passed to a CNN in the second stage to predict the WSI-level label. The experimental results showed that their new encoder was better than CAE and VAE.

6.2.2 Weak-supervised learning-based approaches

A weakly-supervised approach for lung cancer whole-slide image classification was developed in [94]. The method used image-level labels and coarse annotations that pathologists performed by drawing polygons. The method consisted of three parts. The first part was discriminative

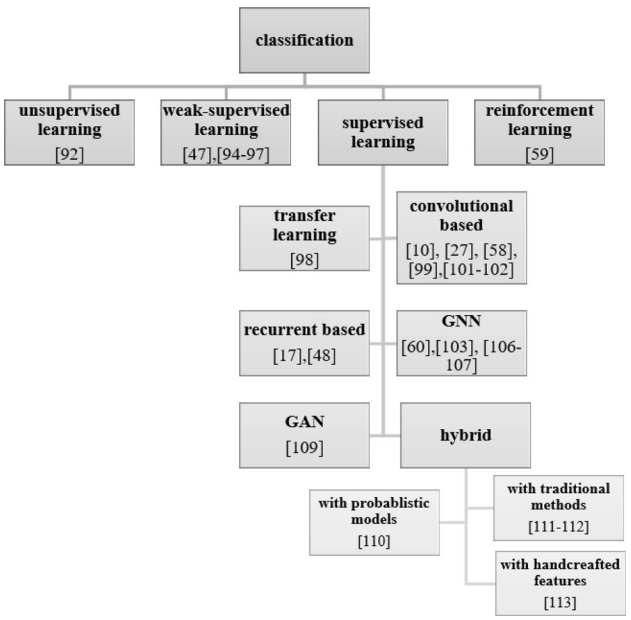


Fig. 7 Categorizing the recent works in the classification task according to the adopted methodology

patch prediction and retrieval using a patch-based CNN model called ScanNet [10]. The second part was a context-aware block selection strategy that used a powerful FCN to examine the obtained spatially contextual information to extract deep representative features. The final part used feature aggregation strategies to generate a global summarizing descriptor for each WSI fed into a Random Forest (RF) classifier for the final WSI-level prediction. Three proposed aggregation methods were used: AvgFeat, WeightedFeat, and MaxFeat. Although the method achieved an accuracy of 82%, it also exhibited multiple limitations; the patch feature embedding collection process used many hyperparameters that needed to be tuned. Moreover, the method was not an end-to-end process; the feature embedding and classification processes were conducted independently, which caused some failure to recognize the ambiguous regions in WSIs.

A two-stage attention-based multiple-instance learning model was proposed in [95]. In the first stage, a binary classification into "cancerous" and "non-cancerous" was performed; tiles were extracted at a 5 × scale and were treated as instances, and the WSI as a bag in a Multi-Layer Perceptron (MLP) model. Then the VGG11 network was used to obtain the instance-level features. The obtained features underwent a dimensionality reduction process using Principal Component Analysis (PCA). After that, the features were clustered to group the semantically similar ones. Next, an attention model was used to determine the most informative and relevant instances. On the other hand, another tile extraction-based clustering was conducted in the second stage at a higher resolution scale (10x). The number of tiles extracted from each cluster was determined using the average attention of the cluster and the overall required tiles numbers. These tiles were fed to the VGG11 network for feature extraction to perform a slide-based classification into benign, low-grade, or high-grade prostate cancer. The model also provided a relevant region visualization by utilizing saliency maps, which improved the results' interpretability. Nevertheless, the used resolutions (5 × and 10x) missed the sufficient high-level information required to capture good features.

The weakly-supervised self-interpretable breast cancer classification method [96] distinguished the WSI images' abnormalities by mining the normal and malignant patches' contrast patterns and generating corresponding probability maps. Normal patches' dominant and discriminative contrast and structural patterns were learned and detected using a fully convolutional autoencoder that minimized the SSIM index. On the other hand, the malignant patches were analyzed by a one-class support vector machine and one-layer neural network and detected implicitly by selecting and distinguishing the patches incompatible with the common normal patterns. After that, a probability map that

Table 7 The details of the recent unsupervised (US), weakly-supervised (WS), and reinforcement learning (LR) based WSI tumor classification approaches

Nature	References	Year	Organ	Method	Preprocessing	Data augmentation	Datasets	Evaluation metric
US	[92]	2018	Breast	CAE: Convolutional AutoEncoders, VAE: Variational AutoEncoders, a novel contrastive training approach	–	Data augmentation as in [100]	Camelyon16 Grand Challenge Dataset	AUC-ROC (0.672)
WS	[94]	2020	Lung	ScanNet + random forest classifier	Irrelevant background removal using OTSU algorithm, then tile extraction of size 244^2	Rotation, translation, flipping, color jittering	Two datasets: Private Lung cancer dataset (SUCC) of 939 WSIs; Public lung cancer WSIs dataset from TCGA	Accuracy (TCGA:0.820, SUCC: 0.973), AUC (TCGA: 0.856)
WS	[95]	2019	Prostate	Attention-based Multi-resolution model	Background removal, color normalization, relevant region extraction using blue ratio image	–	A private two-part dataset: 513 pixel-level annotated prostatectomies tiles and 30 WSIs with slide-level annotations	Accuracy (0.8511)
WS	[96]	2019	Breast	Fully Convolutional Autoencoder	35 patches extraction per processed image of size 256^2 pixels with 30% overlap	Random rotation, Vertical reflection, Horizontal flipping	Breast cancer benchmark biopsy dataset consists of 361 samples: 119 normal tissue, 102 carcinoma in situ, and 140 invasive carcinoma	Accuracy (0.760), F1-score (0.777), DOR (12.876) measure that combines sensitivity & specificity
WS	[97]	2021	Breast, lung	Dual-stream Multiple Instance Learning Network	Non-overlapping patch cropping with size 224^2 at different magnifications, discarding background	Two random image augmentation branches	Two public WSI datasets, including Camelyon16 Grand Challenge Dataset and TCGA lung cancer	Accuracy (Camelyon16: 0.8992, TCGA: 0.9571), AUC (Camelyon16: 0.9165, TCGA: 0.9815), FROC (Camelyon16: 0.4371)
WS	[47]	2021	Breast, colon	C2C: Cluster-to-Conquer	Patch extraction of size 512^2	–	Camelyon16 Grand, private gastrointestinal dataset with 413 WSIs	Accuracy (private dataset: 0.862), Precision (private dataset: 0.855), Recall (private dataset: 0.922), F1-Score (private dataset: 0.887), AUC-ROC (Camelyon16: 0.9112)
RL	[59]	2019	Breast	Deep hybrid attention (CNN + LSTM) based approach	–	Rotation, horizontal and vertical flipping	Public BreakHis Dataset	Patient Recognition Rate (97.5 under 40 × magnification)

Table 8 The details of the recent transfer learning (TL), convolutional based (CB), recurrent based (RB), generative adversarial based (GAB) and graph neural based (GNB) models for WSI tumor classification approaches

Nature	References	Year	Organ	Method	Preprocessing	Data augmentation	Datasets	Evaluation metric
TL	[98]	2021	Breast	Cd-dtffNet: Cross-domain transfer learning and multi-stage feature fusion	–	–	Bridge dataset: BreaKHis, target dataset: private breast histopathological images	Accuracy (0.9827), Sensitivity (0.9764), Specificity (0.9962)
CB	[99]	2018	Breast	Encoder-Decoder networks based	Cropped patches of size 256^2 with an overlap of 56 pixels at different resolutions ($\times$ 5 and $\times$ 10)	Random rotation, horizontal flipping, cropping	240 biopsies were selected from Breast Cancer Surveillance Consortium with 40 $\times$ magnification	F1-score (0.588), mIOU (0.442), Global Pixel Accuracy (0.700)
CB	[10]	2018	Breast	FCN: ScanNet	Removing background, random patch sampling with size 244^2	Translation, rotation, scaling, flipping, color jittering	Camelyon16 Grand Challenge Dataset	FROC (0.8533), AUC (0.9875)
CB	[58]	2019	Breast	Fast ScanNet	Non-informative region removal using the OTSU method	Shifting, rotation, flipping, scaling, color jittering	Camelyon16 Grand Challenge Dataset	FROC (0.8533), AUC (0.9875)
CB	[101]	2019	Stomach	RMDL: Recalibrated Multi-instance Deep Learning	Patch extraction of size 470^2, then sub-patch cropping to size 299^2	Random scaling, random rotation, random flipping	Private Whole Slide Gastric Image dataset (WSGI) contains 408 training and 200 testing images	Accuracy (0.865), Average Classification Score (0.923)
CB	[27]	2019	Stomach, breast	Multistage deep network: feature extractor (GoogLeNet), then segmentation model (U-Net)	Background separation using Otsu threshold, non-overlapping patch cropping of size 256^2	Random cropping, rotation, random left–right flipping, color augmentation	Private Stomach biopsy dataset with $\times$ 20 1,019 WSIs; Camelyon16 and Camelyon17 Datasets	E2E Learning: Accuracy (private dataset: 0.9845, Camelyon16: 0.9814), AUC of the precision-recall curve (PR-AUC) (private dataset: 0.9934, Camelyon16: 0.9924), kappa score (Camelyon17: 76.4)
CB	[102]	2020	Stomach	Deep-Hipo: Multi-scale Receptive Field Deep Learning	Noise removal, background filtering, annotated ROIs localization, non-overlapping patch extraction of size 256^2	–	94 WSIs from two datasets: Stomach Adenocarcinoma (TCGA-STAD) and TCGA Colon Adenocarcinoma (TCGA-COAD)	Patch-based Accuracy (0.937), F1-score (0.934), Precision (0.929), Recall (0.939)

Table 8 (continued)

Nature	References	Year	Organ	Method	Preprocessing	Data augmentation	Datasets	Evaluation metric
GNB	[103]	2019	Kidney	DAG-SVM: Directed Acyclic Graph SVM	Background removal using pixel thresholding, 50% overlapped patch extraction of size 512^2	Random vertical flipping, rotation, noise addition with a probability of 0.5	TCGA with selected 1,584 WSIs contained tumor and 509 normal WSIs	Patch Wise Accuracy (0.9196), Kappa Score (0.794), Micro-Average AUC (0.91), Precision (0.8894), Recall (0.8792)
GNB	[60]	2020	Breast	MS-GWNN: Multi-Scale Graph Wavelet Neural Network	Image resize, color normalization	Flipping, rotation, translation, shear, and linear contrast normalization	BACH 2018 grand challenge dataset, BreakHis dataset	Accuracy (BACH: 0.9375, BreakHis: 0.9967)
GNB	[105]	2020	Lung	Two-stage GNN-MLP framework	–	–	MUSK1 benchmark dataset selected 1026 lung slides from TCGA	Accuracy (0.926), AUC (0.89)
GNB	[106]	2020	Colon	FS-GCN-MIL: deep graph CNN and feature selection, VAE-GAN: variational autoencoder and GAN	Image resize	–	The Cancer Genome Atlas (TCGA) dataset consists of 425 WSIs of Colon Adenocarcinoma	AUC-ROC (0.7102), Accuracy (0.6761), Precision (0.575), Recall (0.7931), F1-score (0.6667)
GNB	[107]	2022	Breast	HACT-Net: Hierarchical Graph Neural Network	Stain normalization	Rotation, mirroring, color augmentations	BACH 2018 dataset, proposed BReAst Cancer Subtyping (BRACS) dataset with 4391 Tumor ROIs from 325 breasts WSIs	Weighted F1-score (7-class: 0.6153, 4-class: 0.6904, 2-class: 0.8415), Accuracy (BACH 4-class: 0.910)
RB	[17]	2018	Breast	CNN based Recurrent Visual Attention Model	Tile extraction of size of 5000^2	–	Camelyon16 Dataset	AUC (0.95), Patch Level and WSI Level Accuracies (0.97 and 0.95), Precision (0.98), Recall (0.82)
RB	[48]	2021	Breast	BiLSTMs	patch extraction of size 256^2	Used but not mentioned	BACH 2018 grand challenge dataset	Accuracy (4-class: 0.900), Class-wise Sensitivity, and Specificity
GAB	[109]	2021	Cervical, lymph node	Selective Synthetic Augmentation with HistoGAN	patch extraction	Proposed selective synthetic augmentation method	Two datasets: cervical dataset; a subset of Patch-Camelyon (PCam) [114] dataset that contains lymph node with metastatic cancer	Accuracy (0.821), AUC (0.881), Sensitivity (0.671), Specificity (0.917)

Table 9 The details of the recent hybrid WSI tumor classification approaches, where HPM (Hybrid with Probabilistic Models), HTM (Hybrid with Traditional Methods), and HHF (Hybrid with Handcrafted Features)

Nature	References	Year	Organ	Method	Preprocessing	Data augmentation	Datasets	Evaluation metric
HPM	[110]	2019	Skin	SuperCRF: Superpixel-based conditional random fields	Tiles splitting of size 2000^2 pixels	–	TCGA dataset contains 105 H&E stained slides with 20 × magnification	Accuracy (0.9648), Precision (0.9644), Recall (0.9629)
HTM	[111]	2018	Breast	Context-aware learning using transferable features	Stain normalization, overlapping patch extraction of size 512^2	Random rotation, flipping	Camelyon16 Grand Challenge Dataset	Accuracy (0.75)
HTM	[112]	2019	Breast	MsC^{2}EF: Multiscale context-cascaded ensemble framework	Stain normalization	–	Public BreaKHis dataset	Image-level Accuracy (0.948)
HHF	[113]	2020	Breast	MF-CNN: Multifaceted fused-CNN with a hybrid-Descriptor	Color normalization, ROI selection, converting RGB image to Blue Ratio image with Otsu's threshold	–	TUPAC16 dataset[a]	F-measure (0.57)

[a]TUPAC16 (TUmor Proliferation Assessment Challenge 2016) Dataset includes the main dataset and two extra datasets. The main dataset contains 821 WSIs with two breast tumor types taken from TCGA, divided as 500 and 321 WSIs for training and testing, respectively. The other two secondary datasets are for the area of interest, with 148 cases, and for mitotic detection, with 73 breast cancer WSIs from different centers

localized the inferred abnormality's locations in the image was generated to provide visualized validation of the results. Moreover, besides the interpretability, another appealing advantage of their proposed method was the generalization since it depended on the normal patches in training; thus, it is not biased toward detecting a specific cancer type. However, this indicated that this method is unsuitable for distinguishing the different malignant tumor categories. Moreover, this method processed the WSI at a low resolution that missed the fine details.

Dubbed Dual-Stream Multiple Instance Learning network (DSMIL) was introduced in [97] that consisted of three components; self-supervised contrastive learning, multiscale attention mechanism, and dual-stream MIL aggregator. The self-supervised contrastive learning extracted good representative features for each WSI patch. Then to further improve the performance, DSMIL incorporated a pyramidal multiscale concatenation strategy that fused feature vectors of each low-magnification patch and all the higher magnification sub-images that contained this patch. Consequently, a trainable distance measurement was adopted to jointly model the patch relations using a patch classifier with the max-pooling operation and the learned features to select the highest-scored patch to generate the final prediction.

The end-to-end Cluster-to-Conquer (C2C) classification framework was developed in [47]. C2C consisted of patch encoding using a convolutional-based model and an adaptive attention-based aggregation approach. C2C began by rearranging each WSI's patches into different clusters to select specific patches in a cluster-based sampling method to feed the ResNet-18 model with diverse discriminative features. Then these feature representations were fed to an adaptive pooling approach with a two-layer fully connected attention module for patch weight calculation to fuse the patch-level representations to produce the final prediction using a classifier with weak supervision. Furthermore, to regularize the intra-cluster attention variance within the single cluster, KL-divergence loss was also used to optimize the clusters' patches' attention weights against the uniform distribution. However, the training process was slow due to clustering, which is considered a time-intensive step.

6.2.3 Reinforcement learning-based approach

A deep hybrid attention approach named "Look, Investigate and Classify" was introduced in [59] for breast cancer classification. As the name suggested, the approach had three stages in each time step; in the "Look" stage, hard

attention was used to crop five patches adaptively from each raw image. Then in the "Investigate" stage, the valuable features were highlighted using a Soft-Attention convolutional-based network (SA-Net). Lastly, the learned features were processed using LSTM in the "classify" stage to generate image classifications and predictions for the next time step. Furthermore, a reinforcement learning approach was adopted to train the whole network because of the non-differentiability in the patch selection process. The model achieved a classification accuracy of 96% using only 15% of raw pixels. However, the patch cropping and classification tasks were trained jointly in a single network, which increased the training difficulty and became less stable.

Table 7 compares the details associated with the different tumor classification approaches reviewed in this section. Some of these approaches adopted the MIL technique.

6.2.4 Supervised learning-based approaches

6.2.4.1 Transfer learning-based approach The Cd-dtffNet [98] incorporated cross-domain transfer learning and multi-stage feature fusion in a breast cancer histopathological classification framework. Using a bridge dataset, the former aimed to reduce the similarity gap between the source and target domains. On the other hand, the latter aimed for simultaneous extraction and adaptive fusion of hierarchical discriminative representative features learned through a novel residual network. Moreover, a regularization mechanism with a penalty was adopted in the fusion strategy to dynamically optimize different multi-level feature weights to achieve both model stability and generalizability. The Cd-dtffNet model suffered from relatively high variance due to a potential mismatch in the scale during the feature fusion, which resulted in instability issues in the back-propagation optimization.

6.2.4.2 Convolutional-based approaches A multi-resolution encoder-decoder architecture for context-aware learning for breast WSI was developed in [99]. Their network incorporated: Input-Aware encoding blocks (IA-RCU), sparsely and densely connected decoding networks, and densely connected fusion blocks. IA-RCU reinjected the input into multiple levels of the encoder to compensate for the spatial information loss caused by the encoder down-sampling operations by reminding the encoder blocks with the input. Moreover, it allowed the direct flow back of gradients into the input paths. The sparsely connected decoding network combined the encoder features aggregated from multiple levels. In contrast, the added skip connections in the densely connected decoding network enhanced the information flow between a decoding block and the same and preceding levels of encoding blocks. Finally, the densely connected fusion block merges the different resolution outputs. In addition, they extended the provided class labels in the used dataset to 8.

A fully convolutional network was developed in [10] called ScanNet. ScanNet was built based on a modified VGG-16 model by transforming the last fully connected layers into convolutional-based ones. ScanNet adopted two training strategies: asynchronous sample prefetching and hard negative mining. The former maximize GPU utilization by using multiple processors. On the other hand, the latter boosts the already trained model performance to increase its discriminative capability by focusing on tricky cases, adding them as false positive samples to the training set, and retraining the model. Moreover, to tackle the sparser probability maps generated as we go deeper in the network because of pooling operations, a dense reconstruction mechanism was designed to integrate the missed views to make the final probability map more precise and denser. Although that ScanNet achieved fast tumor localization, the time cost increased linearly with dense factors.

To speed up the inference time needed to produce a prediction of the breast WSI, Fast ScanNet was introduced in [58]. Fast ScanNet involved using a designed accelerating mechanism called anchor layers to preselect the candidate tumor boundaries for dense scanning of WSI that reduced the redundant computations by sharing them in the overlapping regions to overcome the speed bottleneck issue. Although ScanNet speeds the inference time, the entire area must be scanned for heat map generation during the detection. This can hamper the performance if the computing resources are limited, unlike processing patches with much less size.

A two-stage framework for gastric cancer classification was developed in [101]. In the first stage, discriminative patch selection was performed using a localization network based on an adaptive patch-based fully convolutional version of Inception ResNet-V2. On the other hand, in the second stage, an image-level label prediction using a developed Recalibrated Multi-instance Deep Learning method (RMDL) utilized the discriminative patches obtained from the first stage instead of directly aggregating them. RMDL method involved three modules: local–global instance-wise feature fusion, instance recalibration of the fused features, and multi-instance pooling modules. RMDL explored the contribution of different patches and their interrelationship and dependencies. Accordingly, a recalibration of the feature vector of patches was done based on the importance coefficient learned from the patch's combined local and global features with respect to the image label, which made the prediction more robust. The two stages were trained separately due to the limitations of computational resources.

A two-step structure that combined a patch-based classification model that acted as a local feature extractor with a slide-scale encoder-based segmentation model that considered the global information was introduced in [27]. Furthermore, two training optimization methods were designed to reduce memory consumption; Separate Learning and End-to-End learning. The Separate Learning method was more stable than the End-to-End learning; it first trained the feature extractor model and then used the generated feature vectors in training the segmentation model. On the other hand, End-to-End learning optimized the two models together using a designed strategy to retain the features and gradients between them; this learning method gave more optimal results but consumed longer training times.

A general multi-scale deep learning-based model called Deep-Hipo was developed in [102]. Deep-Hipo utilized the WSI morphological patterns at multiple scales by simultaneously extracting and processing two patches of the same size; a patch from a high magnification level and the other from a low magnification level. Then, two tracks of CNNs with small and large receptive fields were used to process each patch. Deep-Hipo generated the cancer probability heat map by concatenating the last hidden layers of each CNN together.

6.2.4.3 Graph neural-based approaches A Renal Cell Carcinoma classification approach in WSI was introduced in [103]. It combined a deep CNN model with Directed Acyclic Graph SVM (DAG-SVM) method [104] to tackle the imbalance issue of multi-class data. At first, the patches were classified as cancerous or non-cancerous using CNN. The patches with high probabilities were used to create the binary heat map to extract the morphological features needed for survival prediction. After that, another three-class label classification was performed through feature extraction with the same CNN architecture. Then, a subtype classification was done by replacing the last softmax layer of CNN with DAG-SVM by dividing the multi-class classification problem into multiple binary classifications. Two pre-trained networks were chosen as the CNN architecture, ResNet-18 and ResNet-34, fine-tuned on the used data.

The end-to-end Multi-Scale Graph Wavelet Neural Network (MS-GWNN) [60] tackled breast cancer histopathological image classification. It incorporated the spectral graph wavelet localization property in constructing multi-scale structures by providing flexibility in adjusting the node's receptive fields to obtain multilevel tissue structural information in a graph topology. MS-GWNN consisted of three stages: graph construction stage, with nodes representing the discriminative nonoverlapping patches' features generated from modified ResNet-50 and representing the nodes' intrinsic relationships; node classification stage conducted using multiple parallel GWNNs with different scaling parameters to obtain the multi-scale contextual information and consequently multiple node-level predictions; and finally graph-level classification stage using feature aggregation. However, the manual selection process of the appropriate scaling parameter is considered a limitation of MS-GWNN.

In another recent study, an end-to-end two-stage framework for lung WSIs in [105] first sampled the distinct informative patches from a WSI using a color-based method. Then a pre-trained DenseNet was used for feature extraction. After that, these obtained feature vectors were modeled in a fully-connected graph to learn the contextual relations among them using a novel adjacency learning layer that defined and learned the local and global graph nodes' connections in an end-to-end fashion through a sequence of dense layers and cross-correlation. In the second stage, this constructed graph is converted by aggregating these relations into a single summarizing vector representation to perform the classification. An automatic relevancy inference of the patches using attention via graph pooling was used in the condensed-vector generation process along with two fully connected dense layers. They investigated the performance of two graph-based convolutions: ChebNets Convolution based on spectral methods and GraphSAGE Convolution based on spatial methods. The computed attention values and the generated adjacency matrix provided the model the explainability and efficiency.

A framework based on graph convolutional network with MIL method and feature selection strategy (FS-GCN-MIL) [106] was proposed for colorectal cancer histopathological image classification. The proposed framework divided the WSI into non-overlapping patches that are considered instances of a bag. It had three major steps: self-supervised instance-level feature extraction step using a combination of a Variational Autoencoder and Generative Adversarial Network (VAE-GAN) to overcome the lack of instance labels; instance-level feature selection step to obtain the critical relevant features based on the features' discriminating values histogram and using the Maximum Mean Discrepancy (MMD) distance to alleviate the workload; and finally bag-level classification step by employing a Graph Convolutional Network (GCN) to synthesize the selected features and generate the fully trainable bag representation to get the final classification. However, the model is not end-to-end; the preprocessing step of VAE-GAN requires additional expert domain knowledge.

The multi-level entity-graph representation called HierArchical Cell-to-Tissue (HACT) [107] and hierarchical learning (HACT-Net) architecture shifted the analysis

level of histopathological images from pixel-wise to entity-wise. HACT considered the histology image as an entity-graph. In this graph representation, the biological entities such as constituent cells, nuclei, and tissue regions were the nodes, and intra- and inter-entity relationships were the edges. The structure of HACT consisted of three hierarchical levels: a cell-graph representation that characterized low-level cellular morphology and interactions through nuclei detection, nuclei feature extraction, and topology configuration using pre-trained HoVer-Net [108]; a tissue-graph representation that captured the high-level morphology and spatial distribution of tissue microenvironment through tissue regions identification, tissue regions feature representation, and tissue topology construction using SLIC superpixel segmentation, an iterative color-based merging, and superpixel feature extraction using ResNet; and finally, cells-to-tissue graph representation that encoded the relative spatial cell structure within the tissue organization. Various entity sets encoded different morphological and topological information of the image entities to depict tissue compositions at multiple scales were inherently coupled and selected to construct the HACT graph. Afterward, HACT-Net sequentially processed the constructed HACT graph, jointly from fine-level (cells) to coarse-level (tissue regions), to provide a fixed dimensional embedding that described the image to provide mapping to the specific class label.

6.2.4.4 Recurrent-based approaches To adaptively focus on the images' most discriminative regions, an attention-based sequential model was introduced in [17]. It used recurrent neural networks adopted with an attention memory and a description of tissue appearance at different locations in WSI. Two inputs were fed to the recurrent visual attention model: the joint spatial and appearance representations extracted from location and appearance networks. Then, it maintained an internal state that contained information summarizing the previous patch sequence, i.e., it aggregated the extracted contextual information from all individual patches in the image and their corresponding locations. Forty-two highly discriminative patches were selected adaptively at different scales. However, in some cases, the model failed to predict the correct class label even though it correctly identified the discriminative patterns, especially in the micro-metastases case, suggesting that a more complex aggregation strategy may be needed.

An end-to-end classification recurrent-based model was proposed in [48] that used BiLSTMs to capture the forward and backward contextual relationships between the extracted patches features to obtain the image label. The proposed model was robust against the variations in the patch sizes, i.e., it required no constraints about unifying the variable size tumor regions when analyzing and modeling the patch features without risking losing the resolution details. Unlike most of the histopathological classification frameworks proposed in the literature, this model does not follow the traditional two-step approach: classifying independent patches and then aggregating their results to obtain the final diagnostic label; instead, it eventually produces the label by sequentially processing the patches. Thus, the patches were treated as a sequence of the underlying semantic dependencies in the image and extracted using a strategic scanning sampling method that guaranteed the maximal spatial connectivity between them. The sequence was initially formed by accumulating the patch features from each region obtained from pre-trained GoogleNet. Afterward, the sequences were fed to one BiLSTM layer.

6.2.4.5 Generative adversarial-based approach HistoGAN, a synthetic augmentation framework, was developed in [109] based on a conditional GAN model that selectively added some of its generated synthetic patches with high confidence to expand the training dataset. HistoGAN synthesizes the realistic histopathological patches in a coarse-to-fine multi-stage fashion conditioned on class labels. Starting from a noise vector and gradually refining through multiple progressive generations until reaching the required level of quality. Many techniques were used to enhance the quality and stabilize the adversarial training of the network, such as self-attention to capture the internal pixel relationships that held curial information about the color and density; minibatch discrimination to guarantee that the generated examples had some degree of diversification; class conditional batch normalization; and spectral normalization. Moreover, a smoothed version of the Frchet Inception Distance (FID) score was adopted as the selection metric during training to choose the best cGAN model. Nevertheless, they stated that the quality and diversity of the generated synthetic images could be enhanced more.

Table 8 highlights the different characteristics of the reviewed worth-noting recent tumor classification approaches under the fully trained category. The table shows that the increasing trend in tumor classification is that the fully trained end-to-end approaches with the classification approaches based on the neural graph have depicted more attention among them. As in the case of the segmentation task, the models with multi-scale fusion strategies are becoming increasingly popular in the classification approaches due to their effective encoding of discriminative knowledge.

6.2.4.6 Hybrid approaches

- Hybrid with probabilistic models

  A multi-resolution hierarchical classification framework called SuperCRF was developed in [110]. It integrated a developed Spatially Constrained Convolutional Neural Network (SC-CNN) with a superpixel-based machine-learning framework using CRF graphical probabilistic model to improve the classification results. SC-CNN is composed of a Neighboring Ensemble Predictor (NEP) and a conventional CNN to predict the probability of a pixel being in the nucleus center. Eighty-five features were extracted from each superpixel generated using the SLIC algorithm from the low-resolution image and classified using SVM. Consequently, two CRF dependencies were used in SuperCRF: locally based on a cellular level using SC-CNN and globally between the cells and their context using the classified superpixels.
- Hybrid with traditional methods

  A context-aware network for breast cancer classification in histopathological images was introduced in [111]. It had two steps: patch-based classification step and context-aware image-based classification step. A 50-layer ResNet network was used as the patch classifier to transform each image patch into a high-dimensional feature vector, then PCA was used to reduce its dimensionality. After that, these transferable reduced features were given to an SVM classifier with an RBF kernel to aggregate contextual information for WSI to generate the image-wise classification with the aid of a majority voting approach to solve patch overlapping. However, among the multiple available options, the manual selection of the suitable kernel function of the SVM could be tricky.

  For breast histopathological image classification, the Multiscale Context-Cascaded Ensemble Framework (MsC^2EF) [112] consisted of an input layer, four cascaded layers, and a decision layer. In the input layer, a Gaussian spatial pyramid of the image was constructed for coarse- to fine-resolution feature extraction under a single magnification level. This multiscale process ensures diversity; the low-resolution features are less noisy but too sparse, while the high-resolution features provide more details but are noisy and redundant. Furthermore, different operators extracted additional textural features per pyramid layer to quantify the corresponding micro-environment. Next, four ensemble parallel multi-kernel SVMs were stacked for adaptive and iterative contextual feature selection and transfer using a wrapper-based strategy and an adaptive per-layer post-pruning algorithm to eliminate redundant information. To guarantee diversity, four different SVM kernels: Linear, Polynomial, Gaussian, and Sigmoid, were used in MsC^2EF to search four distinctive subspace representations, which were tailored by a wrapper strategy. Finally, in the decision layer, Multi-kernel learning (MKL-SVM) fused the obtained four different feature spaces from the cascaded layers to perform the final classification. The multiscale information in MsC^2EF was obtained from constructing a spatial pyramid from the image rather than using the efficient information from the different magnification levels to overcome the inability to perform one-to-one matching of the ROIs at these multiple magnifications.
- Hybrid with handcrafted features

  A Computer-Assisted Breast Cancer Diagnosis System (CABCDS) was introduced in [113]. The main components of CABCDS were the ROI Selector, Multifaceted Fused-CNN (MF-CNN), Hybrid-Descriptor, and WSI-Scorer. The ROIs were selected from a WSI with low-level magnification based on cell counts, color, and Local Binary Patterns (LBP) texture features. The cell counts were calculated by converting the WSI into a Blue-Ratio image and applying adaptive Otsu's global threshold. Four ROIs were selected, and two actions were performed: forwarding them to MF-CNN and extracting hand-crafted features. MF-CNN utilized information at different granularity by extracting local, intermediate, and global features using three different architectures: CNNseg, CNNdet, and CNNglobal. MF-CNN used transfer learning and deep residual learning to address the class imbalance issue and reduce the complexity. Then the ROI handcrafted features and the dynamic features obtained from MF-CNN were combined to form Hybrid-Descriptor. Each slide's Hybrid-Descriptor was fed to WSI-Scorer, which was a multi-class SVM that used a majority voting to produce the final decision. The selection process to obtain an exhaustive representative set of ROIs is challenging and affects overall performance.

  Table 9 lists the details of the hybrid tumor classification approaches. These recent approaches adopt a combination of multiple feature selection strategies, deep features, and domain-engineered features to improve performance. Although incorporating domain knowledge in the analysis approaches is often performed to guide the deep model, the model's generalizability requirement is disrupted.

6.3 Multi-task learning

Multi-task learning refers to jointly achieving more than one objective through performing multiple tasks in a single framework. Following are summarizations of the recent

works in this category grouped according to their methodologies into unsupervised and weakly-supervised learning-based and supervised learning approaches. The latter is divided into convolutional-based, hybrid approaches with Handcrafted Features and ensemble models. Figure 8 illustrates this categorization with examples of works reviewed in this survey that belong to each category. While Table 10 includes more details and comparisons between them according to their characteristics.

6.3.1 Unsupervised and weak-supervised learning-based approaches

A multi-purpose breast histopathological analysis method was proposed in [115] that explored two use cases: invasive ductal carcinoma detection and lymphoma subtypes classification. It investigated two scenarios that both adopted the residual convolutional autoencoder (FusionNet). The first scenario (Classification by reconstruction) unsupervised trained the FusionNet with sparsity constraints that extracted the relevant latent features important for image reconstruction. On the other hand, the second scenario (Supervised Classification) took only the encoding part of FusionNet and supervised trained it. This approach is referred to as Supervised Encoder FusionNet (SEF). It was observed from the experimental results that the generated false-positive patches in the SEF prediction were sparsely distributed, unlike the ones in the FusionNet prediction that were gathered in certain areas.

Another multi-task approach, called Multi-view Attention-guided Multiple Instance Detection Network (MA-MIDN) [61], tackled the lesion localization and image classification tasks simultaneously by adopting the weakly supervised Multiple Instance Learning (MIL) framework. The end-to-end MA-MIDN framework integrated many algorithms to achieve accuracy and interpretability: multiple-instance neural network, CNN-based feature extractor, novel Multiple-View Attention (MVA) algorithm, and MVA-guided MIL pooling strategy. Each histopathological image was treated as a bag, and its patches as instances in this bag. Then the instance features were extracted using a simple 4-layer CNN network. The developed MVA aimed to learn the different contributions of each instance by computing their attention weights to localize the tumor regions and find ROIs in the image. Two aspects were taken into account to calculate the attention: instance-wise by determining the patch probability belonging to a specific class and bag-wise by determining its influence in classifying the image to a particular class. Then, the proposed MVA-guided MIL pooling strategy aggregated and combined the instance features with respect to their assigned attention weights to obtain discriminative bag-level features by using the permutation-invariant pooling function to obtain the final result bag label. However, the instance-level features generated were relatively simple. Thus, the adoption of more detailed features may enhance the diagnosis performance.

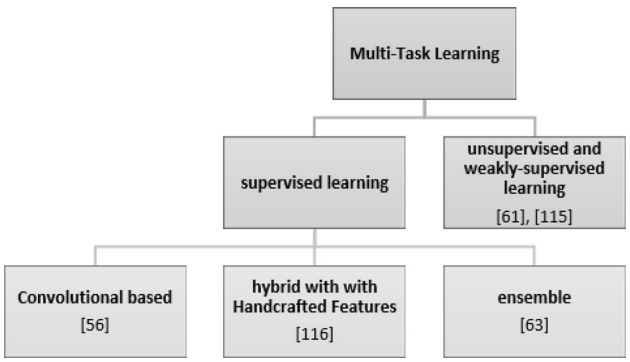


Fig. 8 Categorizing the recent works dealing with multi-task learning according to the adopted methodology

6.3.2 Supervised learning-based approaches

6.3.2.1 Convolutional-based approach To increase the discrimination capability of breast cancer metastasis detection in WSI, a Pyramidal Feature Aggregation ScanNet (PFA-ScanNet) was developed in [56]. PFA-ScanNet fused the local-to-global pyramidal features obtained using diverse receptive fields. The fully convolutional-based PFA-ScanNet consisted of a classification detector and an additional segmentation decoder. Each level in the detector was integrated with a Parameter-efficient Feature Extraction module to extract pyramidal features. Furthermore, Synergistic learning was introduced in PFA-ScanNet to ensure joint simultaneous semantic guided training of both the detector and decoder using a modified truncated version of the binary cross-entropy loss function. Also, they used dense pooling layers in the inference to speed up the scanning process.

6.3.2.2 Hybrid approach with handcrafted features A two-task approach for breast cancer classification in WSI was introduced in [116] for detection and five-class classification. It consisted of three stages. In the first stage, the saliency detection framework detected the salient regions in WSI via a sequential pipeline of four fully convolutional networks that detected the Regions of Interest (ROIs) with localization at multiple magnifications. A tracing log of pathologists' steps when analyzing WSI was used in the training of saliency detection; certain rules were defined that analyze the locations and the time durations of zooming actions of three pathologists when interpreting WSIs to mimic their viewing behavior. The combined results at different levels produced the WSI saliency map.

Table 10 The details of the recent unsupervised (US), convolutional based (CB), hybrid with handcrafted features (HPM), and Ensemble multi-task learning WSI approaches

Nature	Referernces	Year	Organ	Method	Preprocessing/data augmentation	Datasets	Evaluation metric
US	[115]	2019	Breast	SEF: Supervised Encoder FusionNet	Horizontal and vertical flipping, three clock-wise rotations of 90	Two public datasets consisting of 162 WSI and 374 images, respectively	Precision (0.7945), Sensitivity (0.8375), Specificity (0.9177), Accuracy (0.8957), F1-score (0.8154), Balanced Accuracy (0.8776)
US	[61]	2021	Breast	MA-MIDN: Multi-view Attention-guided Multiple Instance Detection Network	Rotation, horizontal flipping, mirror flipping	Three public breast cancer datasets: BreaKHis, BACH, and PUIH [118]	Accuracy (BreaKHis: 0.9633, BACH: 0.9357, PUIH: 0.9276), AUC-ROC (BreaKHis: 0.9983, BACH: 0.9897, PUIH: 0.9851), Precision (BreaKHis: 0.9773, BACH: 0.9618, PUIH: 0.9407), Recall (BreaKHis: 0.9622, BACH: 0.9426, PUIH: 0.9422), F1-score (BreaKHis: 0.9685, BACH: 0.9518, PUIH: 0.9411)
CB	[56]	2019	Breast	PFA-ScanNet: Pyramidal Feature Aggregation ScanNet	Informative tissue regions extractions from WSIs with Otsu algorithm, patch extraction of size 692^2/ Random flipping, scaling, rotation, cropping, color jittering, HSV augmentation	Camelyon16 Dataset, Camelyon17 Dataset	FROC (Camelyon16: 0.902), AUC (Camelyon16: 0.988), kappa Score (Camelyon17: 0.905)
HHF	[116]	2018	Breast	Ensemble of multi-scale FCN's and CNN classifier	Patch extraction	Private 240 digital breast histopathology images	ROC, Accuracy (0.55)
Ensemble	[63]	2020	Lung, Digestive system	Multi-scale U-net fusion model: SU-NET and DU-NET FUSION	Data pre-clustering into three sub-datasets using the k-means algorithm	ACDC@LungHP ISBI 2019 dataset, 200 WSI from DigestPath[a] (MICCAI2019 challenge dataset)	Dice Coefficient (ACDC@LungHP: 0.797, DigestPath: 0.773)

[a]DigestPath: Digestive-System Pathological Detection and Segmentation Challenge

In the second stage, a patch-based five-class fully convolutional network based on these ROIs was used for patch classification. Finally, a fusion of the saliency detection and classification maps was performed in the final stage using a majority voting process to generate a pixel-wise prediction heat map to obtain the final WSI label. The model generated overall slide-level classification accuracy of 55%, while 45 pathologists obtained an average accuracy of around 65%.

6.3.2.3 Ensemble-based approach A fusion-based framework for WSI detection and segmentation was developed in [63] and sub-datasets fusion levels. The fusion model consisted of Shallow U-net (SU-net) and Deep U-net (DU-net). In the multi-scale fusion, three SU-net models were trained on the different resolutions of the whole training dataset. With large samples, the model could detect the tumor areas, but it lacked fine details. Contrarily, models with smaller samples succeed in detecting boundaries with fine details, but they are also more spectacular to noise. On the other hand, in the sub-

datasets fusion, three DU-net models were trained using three sub-datasets created by the *k*-mean algorithm with the same resolution focused on detecting the tumor morphology features. Finally, an overall fusion of all the models was performed and enhanced using the Conditional Random Fields (CRFs) [117] method to generate the segmentation mask.

7 Discussion

The major findings and insights that can be drawn from exploring the reviewed approaches in this survey are discussed next. The numerous histopathological analysis approaches recently proposed in the literature demonstrate the promising results of exploiting deep learning to substitute the traditional analysis approaches that rely on explicit handcrafted features, which have become obsolete. Figure 9 presents the most adopted task and methodology each year, starting from 2018 till 2022, with the most examined organ and experimented public dataset in that year according to the reviewed works in this survey.

7.1 Toward more general CAD systems

As can be depicted from the investigated research in this survey, the recent interest in the past few years is directed toward the segmentation and classification tasks, especially the latter task. The segmentation aims to precisely draw the boundaries that enclose the abnormalities in the medical image to redirect the focus of the medical staff to re-examine these regions to provide more accurate diagnoses and offer better treatment plans. On the other hand, classification aims to determine the existence of these abnormalities in the medical images and their type. Based on that, additional steps like segmentation may be carried out after the classification. Recently, some research works adopted the multi-task learning concept, where their proposed systems aimed to tackle two or more tasks jointly by combining and leveraging the knowledge obtained from these tasks and thus offer a system with more generalization ability. Usually, detection and classification tasks or segmentation and classification tasks are coupled in a single work.

7.2 Toward automated feature learning

The replacement of manually designed features that described the image based on domain expert knowledge with automated features extracted directly from the images makes the deep learning-based methods the ultimate choice for CAD systems, especially with their extraordinary results.

The deep learning-based methods used in histopathological analysis follow multiple categories, amongst them: methods in which the deep model acts as a feature extractor

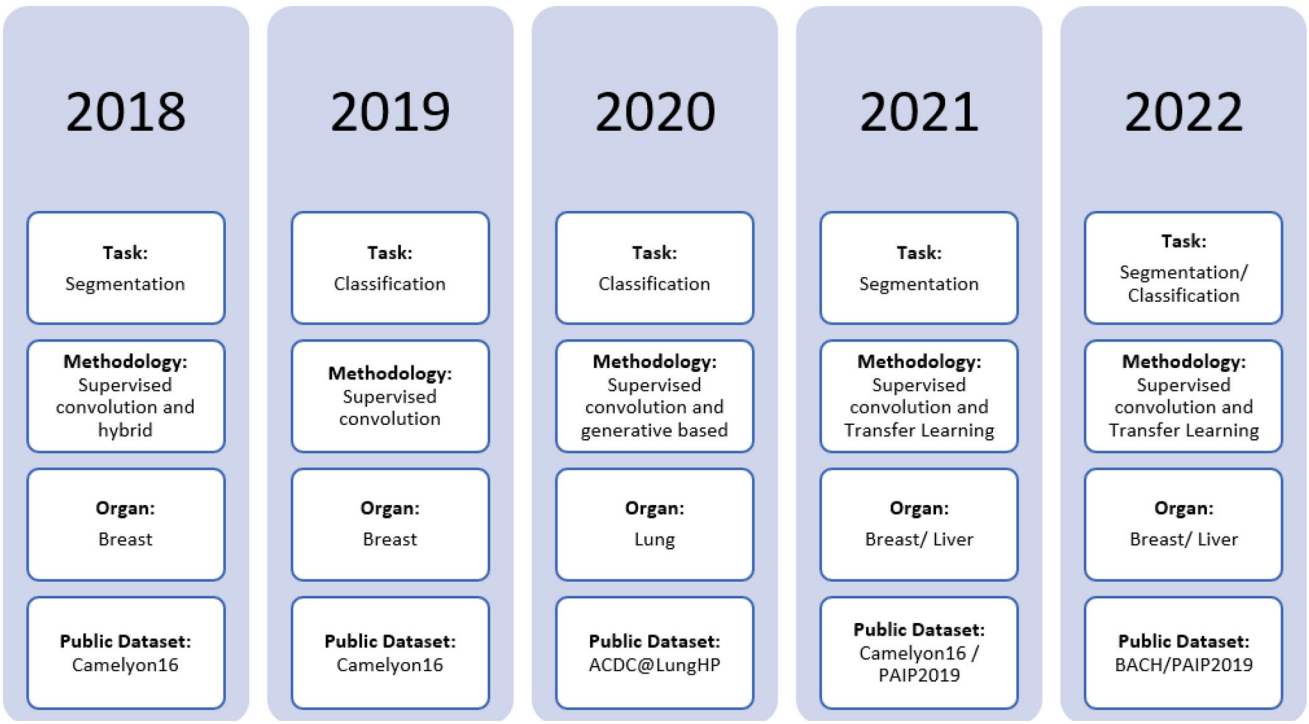


Fig. 9 The most widely used task, methodology, organ and dataset in each year reviewed in this survey

to prepare compact feature vectors that are then passed as input to a traditional or deep.

classifier; methods that are fully trained; methods that used ensemble or cascade of many models to overcome the existing variability in WSIs; methods that utilized the transfer learning concept; methods that are based on semi-, weakly- supervised or unsupervised learning to learn from limited labeled data, data with coarse annotations (slide-level labels) or raw data, respectively.

However, despite the powerfulness of deep learning techniques and the emerging work in this field, there is ripe for further investigation to address the inherent challenges that may face any model that is based on it, such as the lack of the theoretical foundation of the deep learning models; the lack of interpretability of the outcome; and the millions of parameters that must be tuned during the learning procedure to produce good quality results. Furthermore, the fully trained networks that were trained from scratch and exhibited good performance are data-hungry; they require massive well-annotated high-quality histopathological datasets that are very expensive and cumbersome to obtain. Besides, the experienced selection of the model hyperparameters, such as the learning rate, depth, and filter size, is vital but challenging at the same time. Moreover, powerful computational resources are a must to cope up with the large size of WSIs. These factors paved the road toward relying entirely or partially on transfer learning and fine-tuning techniques to transfer knowledge across different domains. However, although transfer learning prevents training the model from using random weights, their large used base datasets, such as ImageNet, contain natural non-medical images, which restrict the obtained benefits from their computed weights since they differ drastically in the associated characteristics.

7.3 Toward dominate architectures

The majority of deep learning-based models for histopathological analysis, as seen from the surveyed papers and despite to which category they belong, are patch-based approaches to address the enormous size of the WSI. They differ in their aggregation strategies and global context encoding methods. However, there is no free lunch! These patch-based approaches have limiting flaws; the neighborhood and global contextual relationships among the patches are not considered since each patch is fed and processed independently and only tackles the information within its small window of view. Consequently, some papers investigated the use of patches of multiple resolutions or used architectures that inherently do this, such as U-Net. Furthermore, using patches of bigger size results in a model with more parameters to train, leading to the previously mentioned computational bottleneck. One more challenge of the patch-based approaches is that analyzing the patches independently violates the pathologist's usual behavior when analyzing these slides.

Most reviewed models preferred to adopt FCN-like architectures, such as U-Net models, rather than using CNN-like architectures because of their shortcomings. For instance, the continuous spatial reduction in CNN caused by the frequent use of the pooling layers [119] results in a loss in the information that may be valuable for representation learning. Furthermore, the popularity of using the U-Net in the models targeted for the segmentation tasks compared with the other FCN models is due to its processing capabilities of propagating the learned contextual information through the different levels in the upsampling decoding path. Also, the usually added skip connections in the U-Net-based segmentation models smooth and refine the prediction results to yield better tumor delineation. On the other hand, the U-Net upsampling path is less critical in the classification models. Adopting and fusing the knowledge obtained from different resolutions of the WSI in a single or ensemble framework prove its effectiveness through the fruitful reported results when injected into U-Net-based models. Nevertheless, the outcomes of these multi-scale models suffer from significant computational time and resources overhead due to processing the WSI patches at multiple magnification levels.

The weakly supervised models are more prevalent in the classification task. They have recently gained more attention from the researchers since no precise tumor boundary delineation is required as in the segmentation task and only needs the coarse image-level labels that are easier to obtain than full pixel annotations. Contrarily, as can be observed from the recently reviewed papers, there are few shy attempts to use the recurrent-based models in tumor analysis due to their sequential processing nature, which is the opposite of the independent analysis of patches usually performed.

Moreover, attention mechanisms are proposed to mimic human visual attention. Their powerfulness is represented by their ability to allow the model in which they are injected, to focus on the discriminative and relevant features in different input positions. This is achieved by assigning a different weight to each input portion according to its importance, then making the output dependent on the weighted average of all the input portions. Different attention mechanisms differ in how these weights are assigned and how the weights are averaged. Therefore, many variants of these mechanisms that vary in complexity have been used in histopathological analysis to boost performance.

GANs play important roles in preprocessing and data augmentation techniques. The strength of GAN relies on its

capability, given a random distribution input, to generate new content with the same distribution as its training data. This is done through two networks that compete against each other, the generator and the discriminator, with the aid of adversarial training. Similarly, Graph Neural Networks (GNNs) are receiving tremendous attention recently because of their end-to-end image-to-graph adaptation that benefits from the expressive power of graphs. Each graph node has a set of features grouped in a vector, while the edges represent the relationships between the nodes and how similar the connected nodes are. In GNN, an iterative message-passing (aka neighborhood aggregation) is performed to enrich the global structural feature learning. However, optimizing, scaling, and training GNN is more complex and resource-demanding than patch-based approaches.

Furthermore, due to their capability of processing the high dimensional images and grouping their pixels into homogenous spatial regions fast and efficiently, superpixels have been used in medical image segmentation in the last decade. This converts the original input image into a much more compact representation that has the ability to preserve the original input properties with comparable performance. Among their important features, one can mention computational efficiency since they convert a large number of pixels into a fewer number of superpixels [104]; robustness to noise [82]; and effective local texture analysis. Although some research papers used superpixels in histopathological analysis field, they are limited in number. The main reason lies in the considerable number of pixels in the WSI that can be considered in millions of pixels; thus, this results in a relatively large number of superpixels to further analysis which puts pressure on the required computational costs.

7.4 Toward trustworthy outcomes

The presence of different challenges and competitions for histopathological analysis boosts the research done in this field by providing well-annotated datasets such as CAMELYON16 and CAMELYON17 grand challenge datasets for breast WSI analysis and PAIP2019 challenge dataset for Liver WSI analysis. This explains the wide research developed for the organs investigated in these challenges, especially under the supervised learning category. Apparently, as observed from the tables, the breast is the most analyzed organ due to the existence of many publicly published datasets and challenges that focus on the histopathological slides extracted from it. However, as can be noticed from the tables, many methods also used private, locally collected WSIs that are not available publicly. This limits these models' reproducibility and thus restricts the comparison capabilities with them or carrying on their research. Another issue depicted in the tables is the lack of unified, standardized evaluation metrics used for any analysis task, which caused those different researchers to use different metrics for the same task, thus complicating the overall evaluation and comparison processes.

The preprocessing steps, such as image resizing, rescaling, normalization, noise removal, or color space conversion, play a vital role in boosting the quality of the generated predictions of the deep learning-based models and alleviating the issue of expensive, unnecessary computation. However, if performed inappropriately, they may cause a loss in image information and thus hinder performance.

7.5 Toward possible avenues to address the challenges in the digital histopathological field

To conclude, the two significant challenges that face the analysis in the histopathological field are the lack of large-scale pixel-wise annotated datasets and imbalanced datasets. These two issues are obligatory for playing an essential role in achieving high-quality predictions and generalizability. However, several techniques and methods can be employed that exhibit the potential to reduce the impact on performance and address the well-known overfitting problem as well:

1. Transfer learning by utilizing the extracted knowledge in two methods: fine-tuning a complex model that was pre-trained previously using large-scale annotated datasets from a different domain or using the pre-trained model as a deep feature extractor to be fed to a shallow or traditional classifier. The latter method raises another challenge: deciding the optimal feature selection criteria, in other words, from which layer to extract the most appropriate deep features.
2. Regularization techniques, among them: weights decay, dropout, and batch normalization.
3. Data augmentation, using conventional transformations or synthetic image generation using GANs, to expand the dataset with respect to two aspects artificially: size and diversity.
4. The use of semi-/weakly- supervised or unsupervised deep learning-based models.

As demonstrated throughout this survey, although the current research works in WSIs for tumor analysis have witnessed remarkable progress and achieved fruitful results, these studies had some limitations, restraining the advancements of recent algorithms and making room for proposing novel tumor analysis models.

8 Conclusion

As can be highlighted from the relatively huge amount of research in the past recent years devoted to tumor analysis using histopathological images and deep learning, deep learning proves its competitiveness in leading the search in the field. This survey tried to enlist notable recent research works in the field of tumor analysis using deep learning models and histopathological images. This survey focused on three analysis tasks: tumor segmentation, tumor classification and multi-tasking. This survey showed that the scope of future research is wide, and the enhancement can be done at many levels; proposing richer and more powerful high-quality datasets that can test the generalization ability of the designed models, i.e., come from different organs, scanners, and medical centers; enhancing and improving the existing deep learning models by addressing the weakness and limitations of the previous models; increasing the power of deep models with the supervision provided by the reinforcement learning that integrates the environmental interactions to enhance the predicted answer, and finally proposing ensemble-based models so that the voting of different participated networks help to address the significant variations in the histopathological slides. Our findings reveal that there is still room to propose a WSI analysis approach that is robust, efficient, less computation demanding, and achieves generalizability, despite the examined organ or the source of WSI, with the aim to overcome the consequences of subjective diagnostic of the pathologist. The goal is to provide a more consistent and accurate tumor diagnosis and the appropriate treatment level for the patient to be applied in real clinical use.

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Declarations

Conflict of interest The authors declare no competing interests.

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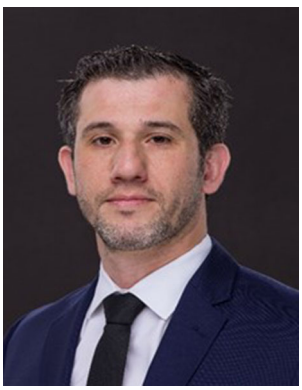
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